

Light curve attitude estimation using particle swarm optimizers

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Abstract

Knowledge of the attitude of a space object is useful in space situational awareness for independently evaluating a satellite's health and characterizing unknown objects. In cases where only non-resolved optical observations are available, the object's attitude may be estimated using a time sequence of brightness observations, also known as the light curve. This attitude estimation problem is plagued with multiple difficulties: even in the absence of noise and when all other relevant factors are perfectly known, the non-uniqueness of the problem means that multiple attitude time histories may fit the light curve equally well. In addition, there is often insufficient information about the object to generate an initial state guess for an estimator. This paper presents a method that estimates an observed object's attitude and angular velocity while accounting for ambiguities and without needing any initial state guess. The only inputs are the light curve, the object's albedo shape, and the object's position relative to the Sun and the observer. The ability of the estimator to resolve attitude time histories is demonstrated using simulated light curves by comparing state estimates against known true states.

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1. Introduction

Space object characterization is a crucial component of space situational awareness. Knowledge of an object's type, shape, reflective properties, and attitude dynamics is useful for identifying the space object, judging its operational state, and predicting its future orbital motion. Optical systems may be used for characterization, but space objects are often too far from an observer for a resolved image to be obtained. In such cases where a resolved image is unavailable, the total brightness of the light from the Sun reflected off the observed object can still be measured. A series of brightness measurements taken over time is called a light curve. Light curves are influenced by the object's shape, material properties, attitude, and position relative

to the Sun and observer. The problem of estimating an object's attitude over time from its light curve is also referred to as the attitude inversion problem.

Light curve inversion to estimate a space object's shape and attitude has been studied for over a century (Russell, 1906; Williams, 1967; Kaasalainen et al., 1992; Kaasalainen and Torppa, 2001a; Kaasalainen and Torppa, 2001b; Kaasalainen et al., 2001; Āurech and Kaasalainen, 2003) but, with some exceptions (Williams, 1967), was initially focused on natural bodies such as asteroids. When analyzing the light curve of a natural body, certain simplifying assumptions can be made, such as that the body is undergoing stable single-axis rotation and its surface is smooth and diffusely reflective (Kaasalainen and Torppa, 2001b; Kaasalainen et al., 2001).

More recently, light curve inversion began to be studied for artificial objects in Earth orbit (Hall et al., 2006; Hall et al., 2007; Wetterer and Jah, 2009; Linares et al., 2014; Benson et al., 2018; Cabrera et al., 2023). However, the

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simplifying assumptions used when investigating natural bodies do not apply to artificial satellites. Satellites are made up of various materials with different reflective properties, their shapes are generally angular rather than smooth, and they are frequently shiny instead of purely diffusely reflecting (Hall et al., 2007). Furthermore, satellites cannot always be assumed to be undergoing stable single-axis rotation (Hall et al., 2007). Even if a satellite is initially spin-stabilized, its rotation will slow due to the presence of eddy currents (Balster et al., 2023), and other effects such as solar radiation pressure and shadowing effects will eventually lead to unstable attitude motion (Benson et al., 2018; Fröh and Jah, 2014). Because of the increased complexity of the light curve inversion problem for artificial objects, it is not uncommon for the shape estimation and attitude estimation components of the inversion process to be examined separately (Hall et al., 2007; Wetterer and Jah, 2009; Fan and Frueh, 2019; Burton and Frueh, 2021; Robinson and Frueh, 2022).

Current attitude inversion methods often require an initial guess for the observed object's attitude and angular velocity (Wetterer and Jah, 2009; Linares et al., 2014; Dianetti, 2020; Cabrera et al., 2023). However, initial guesses are not always available, particularly for objects such as space debris or a tumbling satellite, and many attitude estimation methods can only converge to the correct state if the initial state guess is sufficiently good (Dianetti, 2020). Convergence is particularly sensitive to errors in the initial angular velocity guess, with multiple authors (Wetterer and Jah, 2009; Dianetti and Crassidis, 2023) finding that an unscented Kalman filter (UKF) will only converge for angular velocity errors below 2 deg/s. Using polarized light curve measurements allows convergence for 2 deg/s of angular velocity error, but only if the attitude guess is within 0.6° of the truth (Dianetti and Crassidis, 2023).

If no *a priori* attitude information is available, then another difficulty is encountered in addition to convergence: the ambiguity of the light curve measurements themselves with respect to attitude. Even in the absence of noise and when the observed object's shape, reflective properties, and position relative to the Sun and the observer are perfectly known, each individual measurement in a light curve generally corresponds to many different attitudes (Burton and Frueh, 2020; Clark et al., 2020; Dianetti, 2020; Burton and Frueh, 2021; Gagnon and Crassidis, 2022), and the entire light curve might fit multiple attitude time histories (Clark et al., 2020; Burton and Frueh, 2023; Cabrera et al., 2023; Linares et al., 2014). Methods such as the particle filter (Linares et al., 2014) or adaptive Gaussian filter (Cabrera et al., 2023) are capable of describing the attitude estimate's multimodal probability distribution but still require an initial guess and do not take the entirety of the solution space into account.

One method that does not require an initial state guess has been developed in Clark et al. (2020) and Clark et al. (2022). This method uses a particle swarm optimizer (PSO) to estimate attitude and optical properties from light

curves obtained by space-based observers. A PSO searches the solution space of an optimization problem using particle motion modeled on the social behavior of animals (Eberhart and Kennedy, 1995). The attitude inversion approach developed in these papers (Clark et al., 2022, 2020) uses a PSO to identify possible attitudes at each time step. A first derivative test is used to eliminate impossible transitions between attitudes. However, this method does not take the rotational dynamics of the object into account. Instead, a heuristic limitation is placed on the maximum angular speed and acceleration of the object between measurements. The method has also only been demonstrated for slowly rotating objects. For example, in Clark et al. (2022), a maximum angular speed of 5 deg/s and a maximum angular acceleration of 0.5 deg/s^2 is used.

The method in this study was developed independently of the above method (Clark et al., 2022, 2020) and, unlike that method, fully models torque-free rotational dynamics instead of using a heuristic to restrict the number of possible attitude time histories. Analytic expressions are used to quickly propagate attitude motion, and it is able to resolve attitude time histories even for rapidly tumbling objects, as demonstrated in Section 5. The method in this study uses two PSOs to estimate the attitude and angular velocity of an observed object. The first PSO is a non-social PSO (nsPSO) that locates attitudes that could correspond to one measurement in the light curve. It is called “non-social” because the particles do not communicate with each other, unlike in a standard PSO. The possible attitudes found by the nsPSO are then passed to the second PSO, which uses them as a starting point to search for combined attitude and angular velocity states that fit the light curve as a whole. Only the object's shape, reflective properties, and position relative to the Sun and observer are assumed to be known. The projected use case is for a known piece of space debris or a satellite that is no longer communicating with its operators. Such situations where there is no *a priori* information about the object's attitude and rotational motion are particularly envisioned.

The structure of the remainder of the paper is as follows: Section 2 describes the analytic attitude solutions used for torque-free attitude propagation, then the two PSOs are outlined in Section 3. The phenomenon of “flipped” attitude estimates and how they can be leveraged to generate new estimates is discussed in Section 4. Section 5 discusses the state estimates generated by applying the PSO inversion method to noiseless simulated light curves of both simple and complex objects. This publication expands upon previously published work by the authors (Burton and Frueh, 2023; Burton et al., 2023; Burton et al., 2024). Its primary contributions are a more detailed description of the attitude inversion process outlined in previous publications, new simulated test case results, and a listing of complete analytic attitude expressions for torque-free rotation. These attitude expressions are seldom listed in full due to their length (Fukushima, 2008) and have been adapted specifically to the attitude representation used in this publication.

2. Analytic torque-free motion

2.1. Motivation and attitude representation

The proposed attitude inversion method requires propagating thousands of attitude time histories in parallel, making numeric integration of the rotational equations of motion computationally expensive. Instead, analytic solutions for the torque-free case are used to propagate attitude quickly in the context of a PSO. Note that rotations are approximately torque-free only over short intervals. For long observation windows, the torques acting on the observed object need to be taken into account. The object's attitude may be described using its precession ϕ , nutation θ , and spin ψ angles relative to the angular momentum vector:

$$\mathbf{H} = [\mathbf{I}]\boldsymbol{\omega} \quad (1)$$

where $[\mathbf{I}]$ is the inertia tensor of the object, and the angular velocity of the object $\boldsymbol{\omega} = [\omega_x \ \omega_y \ \omega_z]^T$ is written in the body-fixed coordinate frame $\hat{\mathbf{b}}$. The expressions described in this section apply to any torque-free rotation, including multi-axis tumbling. They may be used for an object with three distinct moments of inertia (MOI) or repeated MOI.

Analytic expressions for the torque-free angular velocity $\boldsymbol{\omega}(t)$ are readily available (Thomson, 1986; Samarasinha and A'Hearn, 1991; Murakami, 2021). However, the expressions for $\phi(t)$, $\theta(t)$, and $\psi(t)$ are not often listed, even in textbooks, likely because of the length and complexity of the expressions (Fukushima, 2008). As such, the torque-free attitude expressions used by the light curve inversion PSO method are fully written out in this section.

Torque-free attitude motion can be divided into short-axis mode (SAM) and long-axis mode (LAM) rotations. Let the body-fixed coordinate system $\hat{\mathbf{b}}$ be a principal axis system such that $I_x \leq I_y \leq I_z$, where I_i is the moment of inertia about $\hat{\mathbf{b}}_i$. SAM and LAM rotations can be distinguished mathematically using the dynamic moment of inertia I_d (Benson et al., 2020):

$$I_d = \frac{H^2}{2T} \quad (2)$$

where H is the magnitude of the angular momentum vector from (1) and T is the rotational kinetic energy:

$$T = \frac{1}{2} \boldsymbol{\omega}^T [\mathbf{I}] \boldsymbol{\omega} \quad (3)$$

The object's angular momentum and kinetic energy are calculated separately for each estimated attitude time history using the estimated angular velocity vector. If $I_d > I_y$, the object is rotating principally about $\hat{\mathbf{b}}_z$ and undergoing SAM rotation. On the other hand, if $I_d < I_y$, the object is undergoing LAM rotation principally about $\hat{\mathbf{b}}_x$. If I_d is equal to any of the moments of inertia I_i , then the object is undergoing a single-axis rotation about $\hat{\mathbf{b}}_i$.

In this study, the attitude of the object is represented using the rotation quaternion $\mathbf{q} = (s, \mathbf{v})$ from a space-fixed inertial frame to the body-fixed frame $\hat{\mathbf{b}}$, where s is the scalar component and $\mathbf{v}$ is the vector component of the quaternion (Vince, 2021). The attitude over time $\mathbf{q}(t)$ is computed as follows:

$$\mathbf{q}(t) = \Delta \mathbf{q}_{\text{SAM/LAM}} \otimes \mathbf{q}_0 \quad (4)$$

where $\mathbf{q}_0$ is the quaternion rotating from the space-fixed frame to the body-fixed frame at the initial time t_0 and $\Delta \mathbf{q}_{\text{SAM/LAM}}$ represents the change in attitude from t_0 . Because of the form of the torque-free attitude solution, the $\Delta \mathbf{q}_{\text{SAM/LAM}}$ unit quaternion is, in turn, related to an Euler angle sequence formed out of the precession ϕ , nutation θ , and spin ψ angles. A 313 (ϕ, θ, ψ) Euler angle sequence is used for SAM rotations, and a 131 (ϕ, θ, ψ) Euler angle sequence is used for LAM rotations.

In the case of torque-free rotation, Euler angles may be safely used to propagate attitude analytically without encountering singularities. For the 313 and 131 sequences, gimbal lock only occurs if $\sin \theta = 0$. However, θ is just the angle between $\mathbf{H}$ and either $\hat{\mathbf{b}}_z$ or $\hat{\mathbf{b}}_x$, meaning that if $\sin \theta = 0$ then the object is undergoing single-axis rotation such that $\boldsymbol{\omega} = \omega \hat{\mathbf{b}}_z$ for SAM or $\boldsymbol{\omega} = \omega \hat{\mathbf{b}}_x$ for LAM. Because no torque is applied to the object, it will never encounter an analytic singularity unless it is initially rotating about only one axis. For objects that start out rotating about a body frame unit vector $\hat{\mathbf{b}}_i$, i.e., $I_d = I_i$, a simplified attitude expression can be used that eliminates the need for Euler angles:

$$\Delta \mathbf{q}_i(t) = \left(\cos\left(\frac{\omega t}{2}\right), -\sin\left(\frac{\omega t}{2}\right) \hat{\mathbf{b}}_i \right) \quad (5)$$

For the general torque-free case, expressions for $\Delta \mathbf{q}_{\text{SAM/LAM}}$ are used that involve the Euler angles ϕ, θ , and ψ and the initial body-frame angular momentum unit vector $\hat{\mathbf{h}} = \mathbf{H}/H$. For SAM rotations, the $\Delta \mathbf{q}_{\text{SAM}}$ quaternion is as follows:

$$\Delta \mathbf{q}_{\text{SAM}}(\phi, \theta, \psi) = \frac{1}{\sqrt{2(h_z+1)}} \left((h_z+1) \cos \frac{\theta}{2} \cos \left(\frac{\phi+\psi}{2} \right) + \sin \frac{\theta}{2} \left(h_y \cos \left(\frac{\phi-\psi}{2} \right) - h_x \sin \left(\frac{\phi-\psi}{2} \right) \right), \right. \\ \left. \begin{bmatrix} -(h_z+1) \sin \frac{\theta}{2} \cos \left(\frac{\phi-\psi}{2} \right) - \cos \frac{\theta}{2} \left(h_x \sin \left(\frac{\phi+\psi}{2} \right) - h_y \cos \left(\frac{\phi+\psi}{2} \right) \right) \\ -(h_z+1) \sin \frac{\theta}{2} \sin \left(\frac{\phi-\psi}{2} \right) - \cos \frac{\theta}{2} \left(h_x \cos \left(\frac{\phi+\psi}{2} \right) + h_y \sin \left(\frac{\phi+\psi}{2} \right) \right) \\ -(h_z+1) \cos \frac{\theta}{2} \sin \left(\frac{\phi+\psi}{2} \right) + \sin \frac{\theta}{2} \left(h_x \cos \left(\frac{\phi-\psi}{2} \right) + h_y \sin \left(\frac{\phi-\psi}{2} \right) \right) \end{bmatrix} \right) \quad (6)$$

For LAM rotations, $\Delta \mathbf{q}_{\text{LAM}}$ is:

$$\Delta \mathbf{q}_{\text{LAM}}(\phi, \theta, \psi) = \frac{1}{\sqrt{2(h_x+1)}} \begin{pmatrix} (h_x+1) \cos \frac{\theta}{2} \cos \left(\frac{\phi+\psi}{2} \right) - \sin \frac{\theta}{2} (h_z \sin \left(\frac{\phi-\psi}{2} \right) + h_y \cos \left(\frac{\phi-\psi}{2} \right)) \\ \left[\begin{aligned} &-(h_x+1) \cos \frac{\theta}{2} \sin \left(\frac{\phi+\psi}{2} \right) + \sin \frac{\theta}{2} (h_z \cos \left(\frac{\phi-\psi}{2} \right) - h_y \sin \left(\frac{\phi-\psi}{2} \right)) \\ &(h_x+1) \sin \frac{\theta}{2} \sin \left(\frac{\phi+\psi}{2} \right) + \cos \frac{\theta}{2} (h_z \cos \left(\frac{\phi+\psi}{2} \right) - h_y \sin \left(\frac{\phi+\psi}{2} \right)) \\ &-(h_x+1) \sin \frac{\theta}{2} \cos \left(\frac{\phi-\psi}{2} \right) - \cos \frac{\theta}{2} (h_y \cos \left(\frac{\phi+\psi}{2} \right) + h_z \sin \left(\frac{\phi+\psi}{2} \right)) \end{aligned} \right] \end{pmatrix} \quad (7)$$

The expressions in (6) and (7) are derived in the Appendix.

2.2. Short-Axis Mode (SAM) Expressions

The analytic expressions for SAM angular velocity $\omega(t)$ are taken from Benson et al. (2020). The same solution is found in Samarasingha and A'Hearn (1991), and similar ones are derived in Thomson (1986) and Murakami (2021):

$$\begin{aligned} \omega_x(t) &= \pm \omega_e \sqrt{\frac{I_d(I_z - I_d)}{I_x(I_z - I_x)}} \text{cn}(\tau, k) \\ \omega_y(t) &= \pm \omega_e \sqrt{\frac{I_d(I_z - I_d)}{I_y(I_z - I_y)}} \text{sn}(\tau, k) \\ \omega_z(t) &= \pm \omega_e \sqrt{\frac{I_d(I_d - I_x)}{I_z(I_z - I_x)}} \text{dn}(\tau, k) \end{aligned} \quad (8)$$

where $\text{sn}(\tau, k)$, $\text{cn}(\tau, k)$, and $\text{dn}(\tau, k)$ are the Jacobi elliptic functions (Abramowitz and Stegun, 1948), and I_d is the dynamic moment of inertia defined in (2). If the body frame $\hat{b}$ is not defined such that $I_x \leq I_y \leq I_z$, then it may be transformed such that the MOI are properly ordered before propagating and transformed back afterward.

The sign of each ω_i component is chosen based on the initial angular velocity $\omega(t_0)$. All terms take the positive sign if $\omega_z(t_0) \geq 0$. If $\omega_z(t_0) < 0$, then ω_z and ω_y take the negative sign unless $\omega_y(t_0) = 0$. In that case, ω_z and ω_x are given the negative sign (Burton and Frueh, 2023).

The constant k is the elliptic modulus, τ is a timelike parameter, and ω_e is the effective spin rate. The constants k and ω_e are calculated as follows (Benson et al., 2020):

$$k^2 = \frac{(I_y - I_x)(I_z - I_d)}{(I_z - I_y)(I_d - I_x)} \quad \omega_e = \frac{2T}{H} \quad (9)$$

And the parameter τ is:

$$\begin{aligned} \tau &= n_\tau(t - t_0) + \tau_0 \\ n_\tau &= \omega_e \sqrt{\frac{I_d(I_d - I_x)(I_z - I_y)}{I_x I_y I_z}} \\ \tau_0 &= \int_0^\Phi \frac{1}{\sqrt{1 - k^2 \sin^2 x}} dx = F(\Phi | k^2) \end{aligned} \quad (10)$$

where $F(\Phi | k^2)$ is the incomplete elliptic integral of the first kind (Abramowitz and Stegun, 1948). The angle Φ in (10) is an angle based on the initial angular velocity $\omega(t_0)$:

$$\Phi = \tan^{-1} \left(\sqrt{\frac{I_y(I_z - I_y)}{I_x(I_z - I_x)}} \frac{\omega_y(t_0)}{\omega_x(t_0)} \right) \quad (11)$$

The analytic expressions for torque-free SAM rotation follow Murakami (2021). The nutation angle $\theta(t)$ and spin angle $\psi(t)$ are found by relating the angular momentum unit vector to the Euler angle sequence. Note that the nutation angle $\theta(t)$ is limited to the range $[0, \pi]$.

$$\theta(t) = \cos^{-1} \left(\frac{I_z \omega_z(t)}{H} \right) \quad (12)$$

$$\psi(t) = \tan^{-1} \left(\frac{I_x \omega_x(t)}{I_y \omega_y(t)} \right) \quad (13)$$

The precession angle $\phi(t)$ is found using the relation between the Euler angle derivative and the body-frame angular velocity (Longuski and Frueh, 2019):

$$\dot{\phi} = \frac{1}{\sin \theta} (\omega_x \sin \psi + \omega_y \cos \psi) \quad (14)$$

After integrating, the solution is found to be (Murakami, 2021):

$$\begin{aligned} \phi(t) &= \frac{H}{I_z} (t - t_0) - \frac{H(I_x - I_z)}{n_\tau I_x I_z} \\ &\times \left[\Pi \left(\text{am}(n_\tau t + \tau_0), -\frac{I_z(I_y - I_x)}{I_x(I_z - I_y)}, k \right) \right. \\ &\left. - \Pi \left(\text{am}(\tau_0), -\frac{I_z(I_y - I_x)}{I_x(I_z - I_y)}, k \right) \right] \end{aligned} \quad (15)$$

where $\text{am}(\tau, k)$ is the Jacobi amplitude function (Abramowitz and Stegun, 1948). The constants k , n_τ , and τ_0 are defined in (9) and (10).

The Π function is the Jacobi incomplete elliptic integral of the third kind (Zill and Carlson, 1970). While Π is useful for the mathematical derivation, the equivalent symmetric elliptic integrals are more convenient when numerically computing the integral (Carlson, 1963; Zill and Carlson, 1970).

2.3. Long-Axis Mode (LAM) Expressions

An alternate set of angular velocity and attitude equations is used if the object is undergoing LAM rotation. The LAM angular velocity functions (Benson et al., 2020) are as follows:

$$\begin{aligned}
\omega_x(t) &= \pm \omega_e \sqrt{\frac{I_d(I_z - I_d)}{I_x(I_z - I_x)}} \operatorname{dn}(\tau, k) \\
\omega_y(t) &= \pm \omega_e \sqrt{\frac{I_d(I_d - I_x)}{I_y(I_y - I_x)}} \operatorname{sn}(\tau, k) \\
\omega_z(t) &= \pm \omega_e \sqrt{\frac{I_d(I_d - I_x)}{I_z(I_z - I_x)}} \operatorname{cn}(\tau, k)
\end{aligned} \quad (16)$$

where the sign of each term is chosen based on the initial angular velocity $\omega(t_0)$. All terms take the positive sign if $\omega_x(t_0) \geq 0$. If $\omega_x(t_0) < 0$, then ω_x and either ω_y or ω_z are given the negative sign. If $\omega_y(t_0) = 0$, then ω_z takes the negative sign; otherwise, ω_y is given it (Burton and Frueh, 2023). The parameter τ is defined as in (10) except that n_τ and τ_0 are given different expressions. The new values for the elliptic modulus k and n_τ are given below:

$$\begin{aligned}
k^2 &= \frac{(I_y - I_z)(I_x - I_d)}{(I_y - I_x)(I_z - I_d)} \\
n_\tau &= \omega_e \sqrt{\frac{I_d(I_z - I_d)(I_y - I_x)}{I_x I_y I_z}}
\end{aligned} \quad (17)$$

and τ_0 is computed as in (10) with Φ for the LAM case now satisfying the following expression:

$$\Phi = \tan^{-1} \left(\sqrt{\frac{I_y(I_y - I_x)}{I_z(I_z - I_x)}} \frac{\omega_y(t_0)}{\omega_z(t_0)} \right) \quad (18)$$

The LAM attitude solution is found in a similar way to the one in Murakami (2021); however, it uses a 131 Euler angle sequence so that ϕ , θ , and ψ are still the precession, nutation, and spin angles, respectively. Because of this change, the expressions are slightly different.

$$\theta(t) = \cos^{-1} \left(\frac{I_x \omega_x(t)}{H} \right) \quad (19)$$

$$\psi(t) = \tan^{-1} \left(\frac{I_z \omega_z(t)}{-I_y \omega_y(t)} \right) \quad (20)$$

A negative sign is included in the denominator of (20) so that the quadrant of $\psi(t)$ can be accurately determined.

The precession angle ϕ is found by integrating the relationship between the precession rate and the angular velocity components:

$$\dot{\phi} = \frac{1}{\sin \theta} (\omega_z \sin \psi - \omega_y \cos \psi) \quad (21)$$

After integration, the $\phi(t)$ expression is found:

$$\begin{aligned}
\phi(t) &= \frac{H}{I_x} (t - t_0) + \frac{H(I_x - I_z)}{n_\tau I_x I_z} \\
&\times \left[\Pi \left(\operatorname{am}(n_\tau t + \tau_0), -\frac{I_x(I_z - I_y)}{I_z(I_y - I_x)}, k^2 \right) \right. \\
&\left. - \Pi \left(\operatorname{am}(\tau_0), -\frac{I_x(I_z - I_y)}{I_z(I_y - I_x)}, k^2 \right) \right]
\end{aligned} \quad (22)$$

where k and n_τ are defined in (17), and τ_0 is found with (10) using Φ from (18). Again, it is recommended that the symmetric elliptic integrals be used to numerically evaluate Π (Carlson, 1963; Zill and Carlson, 1970).

2.4. Initial Euler angles

At the initial time step, $\mathbf{q}(t) = \mathbf{q}_0$, so the quaternion representing the change in orientation, $\Delta \mathbf{q}_{\text{SAM/LAM}}$, must be equal to the identity quaternion per (4). The Euler angles ϕ , θ , and ψ for both the SAM and LAM cases are computed from the body-frame angular momentum unit vector $\hat{h}$, which is computed separately for each estimated attitude time history. The initial angles are found using similar expressions in both the SAM and LAM regimes, so they are discussed together. First, the initial nutation angle θ_0 is computed (Burton and Frueh, 2023).

$$\theta_0 = \begin{cases} \cos^{-1}(h_x) & \text{if } I_d < I_y \\ \cos^{-1}(h_z) & \text{if } I_d > I_y \end{cases} \quad (23)$$

Recall that if $I_d = I_i$, the object is undergoing single-axis rotation about the axis $\hat{b}_i$ and attitude is propagated using (5). Next, the initial spin angle ψ_0 is found that satisfies the following relations (Burton and Frueh, 2023):

$$\begin{cases} \sin \psi_0 = \frac{h_z}{\sqrt{h_x^2 + h_z^2}} & \cos \psi_0 = \frac{-h_y}{\sqrt{h_x^2 + h_z^2}} & \text{if } I_d < I_y \\ \sin \psi_0 = \frac{h_x}{\sqrt{h_x^2 + h_y^2}} & \cos \psi_0 = \frac{h_y}{\sqrt{h_x^2 + h_y^2}} & \text{if } I_d \geq I_y \end{cases} \quad (24)$$

Finally, the initial precession angle ϕ_0 is computed (Burton and Frueh, 2023).

$$\phi_0 = 4 \tan^{-1} \left(\frac{-\sqrt{(\cos \theta_0 + 1)(1 - \cos \psi_0)}}{A + \sqrt{2(\cos \theta_0 + 1)}} \right) \quad (25)$$

where

$$A = \begin{cases} \sqrt{(\cos \theta_0 + 1)(\cos \psi_0 + 1)} & \text{if } \sin \psi_0 \geq 0 \\ -\sqrt{(\cos \theta_0 + 1)(\cos \psi_0 + 1)} & \text{if } \sin \psi_0 < 0 \end{cases} \quad (26)$$

The expressions for the initial Euler angle values in (23), (24), and (25) are derived in the Appendix.

The analytic attitude expressions derived above are used to quickly propagate particles' attitude states in the PSO method described in the following section.

3. Attitude inversion method

The attitude inversion method used in this study is outlined in Fig. 1 (Burton et al., 2023). The first “possible attitude” non-social PSO finds attitudes that could correspond to the light curve measurement at an epoch time. This epoch time is generally the time of the first measurement. The second “full state” PSO searches the space of initial attitudes and angular velocities to find an initial state corresponding to the light curve as a whole. This second PSO uses the possible attitudes found by the first as the starting point for its search and propagates attitude time histories using the attitude expressions derived in Section 2. The estimated states that best fit the measurements are then refined using a quasi-Newton optimizer (Wright, 2006). The only inputs required for the attitude inversion process are the

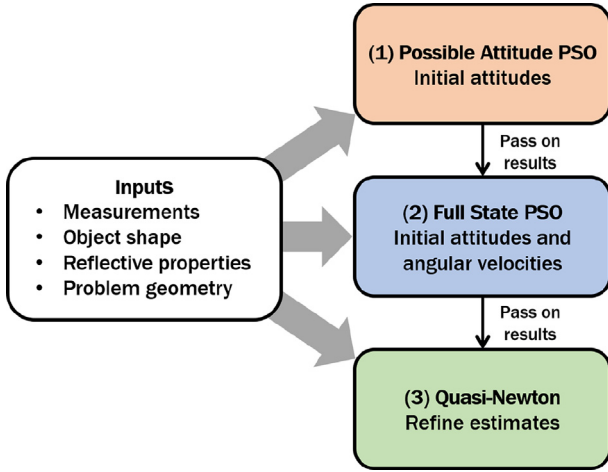


Fig. 1. A flowchart giving an overview of the attitude inversion process (Burton et al., 2023).

light curve measurements, the problem geometry, and the observed object's shape and reflective properties. An initial state guess is not required because of how the PSOs search the solution space. The method makes no assumptions about the object's motion apart from it being approximately torque-free during the observation period.

3.1. Possible attitude non-social particle swarm optimizer

3.1.1. Cost Function

The goal of the first, or “possible attitude,” non-social PSO is to identify possible attitudes of the object at an epoch measurement time. The measured light curve is defined as a vector $\mathbf{z}$ of intensity measurements:

$$\mathbf{z} = [\mathcal{I}_{m,0} \quad \mathcal{I}_{m,1} \quad \dots \quad \mathcal{I}_{m,N-1} \quad \mathcal{I}_{m,N-1}] \quad (27)$$

where N is the number of measurements in the light curve and $\mathcal{I}_{m,j}$ is the intensity measurement observed at time t_j . The time t_j is referred to as the j -th measurement time or the j -th time step. As $\mathbf{z}$ is the measured light curve, it is a function of the true object attitude at each measurement time.

The fit of a particular attitude to the initial measurement is described using the cost function J_{att} (Burton and Frueh, 2023):

$$J_{\text{att}} = \frac{|\mathcal{I}_{m,k} - \mathcal{I}_L(\hat{\mathbf{q}})|}{\mathcal{I}_{m,k}} \quad (28)$$

where $\mathcal{I}_{m,k}$ is the measured light curve intensity at the epoch time step t_k in Watts per meter squared. The epoch time is generally that of the initial measurement, $\mathcal{I}_{m,0}$, but a different epoch time is sometimes chosen to restrict the solution space if many attitudes could correspond to the initial measurement. $\mathcal{I}_L$ is the simulated intensity for a given attitude, and $\hat{\mathbf{q}}$ is the estimated quaternion rotating from a space-fixed coordinate frame, which is taken to be inertial, to a body-fixed frame in the object of interest.

3.1.2. Initialization

The structure of the two PSOs used in this study is outlined in Fig. 2. First, the PSO is initialized with particles (yellow). Second, the particles are moved through the solution space for a set number of iterations (blue). Finally, the best state estimates found by each particle are output (gray).

The initialization phase of a PSO consists of four steps: first, the set of particles is created; second, the cost function at each particle's location is evaluated; third, each particle is assigned the neighbors it will communicate with during the optimization process; finally, the best solutions seen by each particle and its neighbors are stored.

For the possible attitude nsPSO, particles are created in a three-dimensional attitude space. Ultimately, the inversion method outputs solutions in the form of unit quaternions $\hat{\mathbf{q}}_k$ (Vince, 2021). However, the attitude quaternions are internally represented using three angles $\alpha, \beta, \gamma \in [-\pi, \pi]$ that may be converted into a unit quaternion as follows:

$$\hat{\mathbf{q}}_k = (\sin \alpha \sin \beta \sin \gamma, [\cos \alpha \quad \sin \alpha \cos \beta \quad \sin \alpha \sin \beta \cos \gamma]^T) \quad (29)$$

These angles are used by the PSOs instead of the Euler angles described in Section 2 because the estimated initial attitude $\hat{\mathbf{q}}_0$ is distinct from the $\Delta \mathbf{q}$ term as seen in (4) (Burton and Frueh, 2023).

The particles are initialized as evenly spaced in attitude space with coordinates $\hat{\mathbf{x}}_i = [\alpha_i \quad \beta_i \quad \gamma_i]^T$, where i is the particle's index. Each particle is also assigned a random initial velocity $\mathbf{v}_i = [v_{i,\alpha} \quad v_{i,\beta} \quad v_{i,\gamma}]^T$ in the solution space. This step corresponds to Block (1) in Fig. 2.

The initialization process continues with the evaluation of the cost function J_{att} at each particle's location $\hat{\mathbf{x}}_i$. Each particle stores its current position and the corresponding cost as the best solution $\hat{\mathbf{x}}_{B,i}$ it has seen with cost $J_{B,i}$.

The PSO may also allow particles to communicate with each other. If this is the case, then in Block (3), each particle is assigned other particles as “neighbors,” and the lowest-cost solution seen by each particle and its neighbors is stored as its local best $\hat{\mathbf{x}}_{l,i}$ with cost $J_{l,i}$. The possible attitude nsPSO does not use neighbors for reasons discussed in Section 3.1.3. The assignment of neighbors in the full state PSO is discussed in Section 3.2.2. The best solution seen by any particle in the swarm may also be recorded as the globally best solution $\hat{\mathbf{x}}_g$ with cost J_g . Allowing particles to be informed by what other swarm members have seen makes it more likely that the swarm will converge on a global minimum rather than many local minima.

3.1.3. Iterations

After initializing the PSO, the particles are moved through solution space a set number of times to look for the minima of J_{att} in Blocks (5) through (8) of the flowchart in Fig. 2. Before moving the particles, their velocities are updated based on the best solutions found so far.

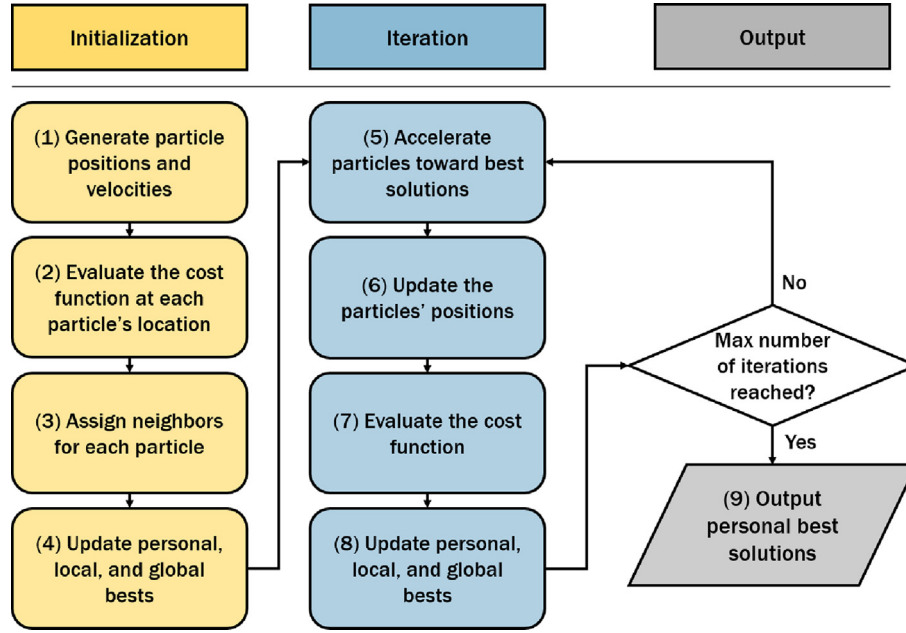


Fig. 2. A flowchart of the structure of the two PSOs used in the attitude inversion method.

In Block (5), each particle is accelerated by a random amount according to the following expression:

$$\mathbf{v}_i^+ = \mathbf{v}_i^- + c_v [R_b(\hat{\mathbf{x}}_{B,i} - \hat{\mathbf{x}}_i) + w_l R_l(\hat{\mathbf{x}}_{l,i} - \hat{\mathbf{x}}_i) + w_g R_g(\hat{\mathbf{x}}_g - \hat{\mathbf{x}}_i)],$$

$$R_x \sim U(0, 1) \quad (30)$$

where $\mathbf{v}_i^{-/+}$ is the velocity of the i -th particle before and after the update, c_v is a positive constant, and the R_x terms are random numbers between 0 and 1. The non-negative weights w_l and w_g control how much importance is given to the local best and globally best solutions, respectively. The magnitude of $\mathbf{v}_i^+$ is capped at a set value $v_{\max}$.

For the possible attitude non-social PSO, w_l and w_g are set to zero. As a result, the acceleration expression in (30) is rewritten as follows:

$$\mathbf{v}_i^+ = \mathbf{v}_i^- + c_v [R_b(\hat{\mathbf{x}}_{B,i} - \hat{\mathbf{x}}_i) + \delta \mathbf{v}], \quad R_b \sim U(0, 1) \quad (31)$$

where $\delta \mathbf{v}$ is a random perturbing vector used to ensure that the particles search more of the solution space than they otherwise would. The components of $\delta \mathbf{v}$ are uniformly distributed, and their maximum magnitude is scaled to the distance between the particle's current location $\hat{\mathbf{x}}_i$ and the location of the best solution it has seen $\hat{\mathbf{x}}_{B,i}$.

These modifications force each particle to consider only the best solutions it has personally encountered, removing the communication between particles that is usually characteristic of PSOs. It is still referred to as a PSO because it is built using that framework, but it is a “non-social” variation of the method.

The lack of communication between particles means that the possible attitude nsPSO will find a larger number of local minima than it otherwise would. In many contexts, this would not be desirable behavior. However, the possible attitude nsPSO aims to locate as many attitudes as

possible, so particles settling into many local minima is desirable in this context.

Once the particle velocities have been updated, the particles are moved through the solution space in Block (6) using the expression shown below:

$$\hat{\mathbf{x}}_i^+ = \hat{\mathbf{x}}_i^- + \mathbf{v}_i^+ \quad (32)$$

where $\hat{\mathbf{x}}_i^{-/+}$ is the position of the i -th particle before and after being updated.

Next, the cost function J_{att} is evaluated at each particle's new location in Block (7). The particles' personal and local best solutions and the swarm's global best solution are then updated in Block (8). A particle's neighbors do not change after they are initially assigned in Block (3). If the PSO has run for the desired number of iterations, it outputs the personal best solutions $\hat{\mathbf{x}}_{i,B}$ found by each particle. If not, the PSO returns to Block (5), and the particles are accelerated and moved again per (30) and (32).

3.1.4. Motion in angle space

The definition of the attitude space representation in (29) is periodic in terms of the three angles $\{\alpha, \beta, \gamma\}$. The position update expression (32) allows unbounded movement of the particles through solution space, which may result in two particles converging to the same attitude quaternion but being widely separated in attitude space. If those particles are neighbors in the full state PSO, discussed below in Section 3.2, such a situation may cause both particles to accelerate away from their common solution. Therefore, the periodic nature of $\hat{\mathbf{q}}(\alpha, \beta, \gamma)$ in (29) is leveraged to keep the particles within the range of $-\pi$ to π . This modification to the position update is also implemented for the possible attitude nsPSO to keep the internal attitude representation consistent through all steps of the inversion process.

Let $\xi \in \{\alpha, \beta, \gamma\}$ be one of the angles that map onto an attitude quaternion according to (29). The periodicity of the attitude representation means that changing ξ by $\pm 2\pi$ results in the same attitude representation. The addition and subtraction operations in (30) and (32) are therefore calculated as follows (Burton and Frueh, 2023):

$$\xi_A \pm \xi_B = \begin{cases} \xi_A \pm \xi_B - 2\pi & \text{if } (\xi_A \pm \xi_B) > \pi \\ \xi_A \pm \xi_B + 2\pi & \text{if } (\xi_A \pm \xi_B) < -\pi \\ \xi_A \pm \xi_B & \text{otherwise} \end{cases} \quad (33)$$

As a result of these operations, whenever a particle's location is updated, any angle coordinates that would fall outside the allowed range are shifted by 2π so that they fall inside it.

3.2. Full state particle swarm optimizer

3.2.1. State and cost function

Once a set of possible attitudes is found at the epoch measurement time, a second “full state” PSO is used to find epoch time attitude and angular velocity states $\hat{\mathbf{y}}_i$ matching the measured light curve (Burton and Frueh, 2023):

$$\hat{\mathbf{y}}_i := \begin{bmatrix} \hat{\mathbf{x}}_i \\ \hat{\boldsymbol{\omega}}_i \end{bmatrix} \hat{\mathbf{x}}_i := \begin{bmatrix} \alpha_i \\ \beta_i \\ \gamma_i \end{bmatrix} \hat{\boldsymbol{\omega}}_i = \begin{bmatrix} \omega_{i,x} \\ \omega_{i,y} \\ \omega_{i,z} \end{bmatrix} \quad (34)$$

where α_i, β_i , and γ_i correspond to the same angles in (29), and $\omega_{i,x}, \omega_{i,y}$, and $\omega_{i,z}$ are the components of the object's estimated angular velocity $\hat{\boldsymbol{\omega}}_i$.

The cost function J_{LC} of the full state inversion PSO is based on how well a propagated light curve $\hat{\mathbf{z}}_i$ matches the measured light curve $\mathbf{z}$. This cost function is distinct from the possible attitude nsPSO cost function (28), which only compares the two light curves at the epoch time step (Burton and Frueh, 2023).

$$J_{LC}(\hat{\mathbf{y}}_i) = \frac{1}{\|\mathbf{z}\|^2} \sum_{j=0}^{N-1} (\mathcal{J}_{m,j} - \mathcal{J}_L(\hat{\mathbf{q}}_j))^2 = \frac{\|\mathbf{z} - \hat{\mathbf{z}}_i\|^2}{\|\mathbf{z}\|^2} \quad (35)$$

where N is the number of measurements in the input light curve, $\mathcal{J}_{m,j}$ is the measured light curve intensity at time t_j , and $\hat{\mathbf{q}}_j$ is the attitude at t_j found by propagating the epoch state $\hat{\mathbf{y}}_i$ using (12), (13), and (15) for SAM rotations, or (19), (20), and (22) for LAM rotations. The rotational mode is determined for each particle individually using the dynamic moment of inertia, I_d , for its estimate as defined in (2). Therefore, for a given iteration, the full state PSO generally propagates attitude time histories using the SAM expressions for some particles and the LAM expressions for others.

Because the cost J_{LC} increases quadratically, a linear error E_{frac} is used to report how well a state estimate matches the light curve to make comparing multiple estimates more intuitive for human operators. E_{frac} is the ratio between the magnitude of the difference between the estimate's light curve and the measured light curve, divided

by the magnitude of the measurement vector (Burton et al., 2024):

$$E_{\text{frac}}(\hat{\mathbf{y}}_i) = \sqrt{J_{LC}} = \frac{\|\mathbf{z} - \hat{\mathbf{z}}_i\|}{\|\mathbf{z}\|} \quad (36)$$

where J_{LC} is defined as in (35) and $\mathbf{z}$ is the vector of light curve measurements.

3.2.2. Full State PSO particle creation

The states $\hat{\mathbf{y}}_i$ in (34) are divided into two kinds: the attitude states $\hat{\mathbf{x}}_i$, and the angular velocity states $\hat{\boldsymbol{\omega}}_i$. When initializing the PSO, the attitude and angular velocity states are generated separately, and then a particle is initialized for each attitude-angular-velocity pairing.

The attitude states are the possible attitudes found using the possible attitude nsPSO, and the angular velocity states are selected with a fixed maximum magnitude that is generally equal to the Nyquist frequency of the input light curve. These angular velocity states may be either randomly generated as in Burton and Frueh (2023) or distributed approximately evenly within the sphere (Burton et al., 2024). Assuming that the light curve measurements are Δt seconds apart, then this maximum magnitude is as follows (Burton and Frueh, 2023):

$$\omega_{\text{max}} = \frac{\pi}{\Delta t} \text{ rad/s} \quad (37)$$

The maximum magnitude may also be set manually.

Once n initial attitudes $\hat{\mathbf{x}}_{(k)}$ and m angular velocities $\hat{\boldsymbol{\omega}}_{(k)}$ have been generated, they are combined to form the set of nm initial state guesses $\hat{\mathbf{y}}_i$ where:

$$\begin{aligned} & \{\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2, \dots, \hat{\mathbf{y}}_m, \hat{\mathbf{y}}_{m+1}, \dots, \hat{\mathbf{y}}_{nm}\} \\ &= \left\{ \begin{bmatrix} \hat{\mathbf{x}}_{(1)} \\ \hat{\boldsymbol{\omega}}_{(1)} \end{bmatrix}, \begin{bmatrix} \hat{\mathbf{x}}_{(1)} \\ \hat{\boldsymbol{\omega}}_{(2)} \end{bmatrix}, \dots, \begin{bmatrix} \hat{\mathbf{x}}_{(1)} \\ \hat{\boldsymbol{\omega}}_{(m)} \end{bmatrix}, \begin{bmatrix} \hat{\mathbf{x}}_{(2)} \\ \hat{\boldsymbol{\omega}}_{(1)} \end{bmatrix}, \dots, \begin{bmatrix} \hat{\mathbf{x}}_{(n)} \\ \hat{\boldsymbol{\omega}}_{(m)} \end{bmatrix} \right\} \end{aligned} \quad (38)$$

The initial velocities of the particles are then randomly generated. The velocities of each particle in the attitude state space $\mathbf{v}_{i,x}$ and angular velocity state space $\mathbf{v}_{i,\omega}$ are generated and updated separately.

For the full state PSO, the local weight w_l and global weight w_g in (30) are not necessarily set to zero. Therefore, particles are assigned neighbors at initialization, which are used to obtain the $\hat{\mathbf{x}}_{l,i}$ solutions. Each particle's neighbors are the n_l particles with the same initial attitude estimate $\hat{\mathbf{x}}_i$ and the angular velocities closest to $\hat{\boldsymbol{\omega}}_i$. Therefore, the particle $\hat{\mathbf{y}}_i = [\hat{\mathbf{x}}_i \ \hat{\boldsymbol{\omega}}_i]^T$ may be assigned $\hat{\mathbf{y}}_j = [\hat{\mathbf{x}}_j \ \hat{\boldsymbol{\omega}}_j]^T$ as a neighbor if two conditions are met. First, $\hat{\mathbf{x}}_i$ must equal $\hat{\mathbf{x}}_j$. Second, the angular velocity distance $\|\hat{\boldsymbol{\omega}}_i - \hat{\boldsymbol{\omega}}_j\|$ must be among the lowest n_l angular velocity distances between $\hat{\mathbf{y}}_i$ and all particles meeting the first condition. A particle's neighbors do not change after they have been initialized.

3.2.3. Motion in the solution spaces

After initialization, the full state PSO is run as described in Section 3.1.3. However, because the attitude and angular

velocity estimate spaces have different properties, the state components $\hat{\mathbf{x}}_i$ and $\hat{\omega}_i$ are updated separately.

The angles representing the attitude quaternion are bounded on $[-\pi, \pi]$ just as for the possible attitude nsPSO. Therefore, the differences between a particle's attitude estimate $\hat{\mathbf{x}}_i$ and its best solutions $\hat{\mathbf{x}}_{B/l/g,i}$ are calculated using (33). The position of a particle in attitude space is also updated according to (33) to ensure the angle coordinates stay in the allowed range.

The angular velocity space is treated as a Cartesian space and updated using (30) and (32). Any particles whose angular velocity estimate magnitude exceeds $\omega_{\max}$ from (37) have their velocities $\mathbf{v}_{i,\omega}$ reversed and are moved so that they lie on the sphere with radius $\omega_{\max}$.

Note that the attitude and angular velocity estimates may use different maximum velocities $v_{\max,x}$ and $v_{\max,\omega}$, and acceleration constants $c_{v,x}$ and $c_{v,\omega}$ when updating a particle's velocity in their respective spaces using (30).

4. The flip problem

Symmetries in the propagation of light (Hapke, 2012; Minnaert, 1941) significantly impact the results found by any attitude inversion method. The quantity $\mathcal{I}_m(\hat{\mathbf{u}}, \hat{\mathbf{s}})$ is defined as the light measured from a uniform surface by an observer in the direction $\hat{\mathbf{u}}$ given a light source in the direction $\hat{\mathbf{s}}$. The symmetries in the propagation of light mean that the same amount of light will be seen if the light source and observer directions are swapped.

Furthermore, assuming torque-free rotation and that the relative positions of the observer and Sun remain constant relative to the observer, then there is always at least one attitude time history that produces identical measurements to the true attitude motion. This attitude time history is found by rotating the object 180° about the bisector of $\hat{\mathbf{u}}$ and $\hat{\mathbf{s}}$ so that the two unit vectors are “flipped.” Because torque-free attitude motion expressions in Section 2 only depend on the initial angular momentum vector, the two unit vectors remain “flipped” at all times, resulting in an unchanged light curve.

This phenomenon explains the attitude estimates 180° from the truth identified in previously published research on the PSO attitude inversion method (Burton and Frueh, 2023). As discussed in Section 5, the truth may be distinguished from the “flipped” estimate by taking another set of measurements with a different problem geometry. In addition, to ensure that more of the solution space is searched when the best estimates from the full state PSO are refined, the “flipped” counterparts of the estimates are also refined.

5. Results

The PSO attitude inversion method is demonstrated by estimating the attitude of space objects using noiseless simulated light curves. Each light curve consists of 25 simu-

lated measurements taken over 40 s, corresponding to a Nyquist frequency of 1.885 rad/s. For a set number of measurements, the Nyquist frequency is coupled to the observation window's length and puts a constraint on the maximum angular velocity that can be resolved. The set of angular velocities used to initialize the full state PSO is evenly spaced in a sphere with a radius equal to the Nyquist frequency. The spacing between angular velocity guesses is controlled by the parameter ω_{space} .

The positions of the object, observer, and Sun are assumed to be perfectly known, along with the object's shape and reflective properties. For each case, the observed object is in a circular geostationary orbit over the 106° W line of longitude (semi-major axis 42,241 km, eccentricity 0, inclination 0). The inversion method may be used in any orbital regime. However, objects in lower orbits experience more external torques, so their rotational motion can be approximated as torque-free only over short measurement windows. The measurements are taken by a ground-based observer at 106° W, 45° N.

In each case, the object's total angular velocity magnitude is 1.5 rad/s. This angular velocity magnitude is chosen to demonstrate the inversion process's ability to resolve the attitude of rapidly rotating objects. Pieces of space debris have been observed with comparable apparent rotation rates (Schildknecht et al., 2015).

Total run times are provided along with the results for each test case. The test cases are run and timed on a Windows laptop with a 4-core 2.2 GHz processor and 8 GB of RAM. The exact run times vary depending on what other processes are being run and whether or not the laptop is plugged in. As a result, the times provided assume that the laptop is plugged in and other memory-intensive processes are not being run.

In Case 1, a cylindrical object reflecting light purely diffusely is observed. In Case 2, a regular tetrahedron reflecting light according to a Cook-Torrance model is observed. Case 3 uses a model of the Landsat 8 satellite (Meaney, 2016) instead of the regular tetrahedron. A second observer is also added that takes 25 additional measurements offset 2.2 s from the originals. The attitude time history of the Landsat 8 model is identical to that of the tetrahedron in Case 2. These differences are summarized in Table 1.

The light curve measurements are simulated in different ways for the three objects. For the regular cylinder, the light reflecting off of each surface is calculated directly using a Lambertian reflection model (Robinson and Frueh, 2022). Since the reflection model assumes that the object is made of flat facets, the cylinder's curved sides are approximated as being made up of 32 rectangular faces. For the tetrahedron, a Cook-Torrance reflection model (Duvenhage et al., 2013) with a diffuse reflection coefficient of 0.8, a specular coefficient of 0.2, and a roughness parameter of 10 is used for all surfaces. The same reflection model is used for the Landsat 8 model. In order to model reflections quickly for the more complex satellite, however, the

Table 1

A summary of the setup for each test case.

Case	Object	Observers	Light Curves	Offset	Measurements
1	Cylinder	1	1	—	25
2	Tetrahedron	1	1	—	25
3	Landsat 8	2	2	2.2 s	50

measurements used in Case 3 are generated using a neural network model (Pedregosa et al., 2011) trained using high-fidelity simulated measurements (Robinson and Frueh, 2022).

Portions of the discussion accompanying these results were originally published in Burton et al. (2024).

5.1. Particle swarm optimizer parameters

The parameters used in the possible attitude nsPSO for each test are given in Table 2. The possible attitude nsPSO outputs all the personal best attitudes $\hat{x}_{B,i}$ found by particles with a cost J_{att} below the threshold J_{thresh} according to (28). The maximum velocity v_{max} and acceleration constant c_v from (31) are constant across all three test cases.

The parameters used in the full state PSO are shown in Table 3. In all three test cases, each particle is assigned twelve neighbors. For Cases 1 and 2, the full state PSO runs for 50 iterations; for Case 3, it runs for 100 iterations. The extra iterations are offset by halving the velocity $v_{\text{max},\omega}$ and acceleration $c_{v,\omega}$ of the particles in angular velocity space. These settings result in a higher-resolution search of the solution space in Case 3. The local weight w_l in (30) is 0.5, and the global weight w_g is zero for all test cases. The maximum velocity $\hat{v}_{\text{max},v}$ and acceleration constant $c_{v,x}$ of the particles in attitude space are the same across all test

Table 2

The parameters used to run the possible attitude nsPSO for each test case.

	Case 1	Case 2	Case 3
J_{thresh}	10^{-3}	10^{-3}	10^{-3}
Particles	17,576	17,576	314,432
Iterations	10	10	10
v_{max} [rad]	0.05	0.05	0.05
c_v [rad]	0.2	0.2	0.2

Table 3

The parameters used to run the full state PSO for each test case.

	Case 1	Case 2	Case 3
Neighbors	12	12	12
Iterations	50	50	100
w_l	0.5	0.5	0.5
w_g	0	0	0
$v_{\text{max},x}$ [rad]	0.05	0.05	0.05
$c_{v,x}$ [rad]	0.1	0.1	0.1
$v_{\text{max},\omega}$ [rad/s]	0.2	0.2	0.1
$c_{v,\omega}$ [rad/s]	0.4	0.4	0.2
ω_{space} [rad/s]	0.35	0.45	0.45

cases. The spacing between angular velocity guesses, ω_{space} , is smaller for Case 1 than for Cases 2 and 3. The spacing parameter controls how many angular velocity guesses are generated and thereby controls the number of particles used by the full state PSO. The spacing parameter and the possible attitude nsPSO settings are iteratively tuned until attitude estimates with error fractions on the order of 10^{-3} or lower are located. For well-behaved test cases, such as Case 1 and Case 2, even lower error fractions may be sought.

5.2. Case 1: Spinning GEO cylinder with one observer

The light curve analyzed in Case 1 is produced by a diffusely-reflecting cylinder (Robinson and Frueh, 2022) that is meant to approximate a rocket upper stage tumbling end over end. The cylinder has a diameter of 3.05 m and a height of 7 m. The cylinder's moments of inertia are calculated by assuming a 2000 kg mass that is unevenly distributed through the cylinder's volume: $I_{x,\text{cyl}} = I_{z,\text{cyl}} = 4923.2 \text{ kg} \cdot \text{m}^2$, $I_{y,\text{cyl}} = 2325.6 \text{ kg} \cdot \text{m}^2$.

The cylinder is shown in Fig. 3a, and as discussed above, the sides are made up of 32 rectangular faces for the purposes of simulating reflections. The cylinder's surfaces all have a diffuse reflection coefficient of 0.8.

The initial attitude of the cylinder is chosen such that its axis of symmetry is 30° from the inertial x-axis, and the angular velocity ω_0 is primarily along the body-fixed z-axis to induce an end-over-end tumble.

$$\begin{aligned} q_0 &= (0.2588, [0 \ 0 \ 0.9659]^T) \\ \omega_0 &= [0.2000 \ 0.2000 \ 1.4731]^T \text{ rad/s} \end{aligned} \quad (39)$$

The observed light curve is shown in Fig. 3b. The light curve measurements correspond to the true attitude time history in Fig. 4a, which shows the components of the true attitude quaternion q over time. The angular velocity components are shown in Fig. 4b. As expected for an axisymmetric object, the angular velocity component along the axis of symmetry remains constant, while the transverse components vary over time.

Using the light curve measurements, the possible attitude nsPSO finds 97 possible initial attitudes after running for ten iterations with 2,744 particles. Each iteration of the possible attitude nsPSO takes about 0.2 s. The full state PSO is initialized using these possible attitudes, resulting in 74,593 particles. After running for 50 iterations, the best 250 solutions output by the full state PSO and their “flipped” counterparts are refined using the quasi-Newton method. The complete inversion process takes about 6.5 h to run.

The five estimates which best fit the input measurements are shown in Table 4. The error E_{frac} in the estimated light curves is defined as in (36), and the average errors in the estimated direction of the axis of symmetry and angular momentum vector are calculated using the difference between the true and estimated object state at each

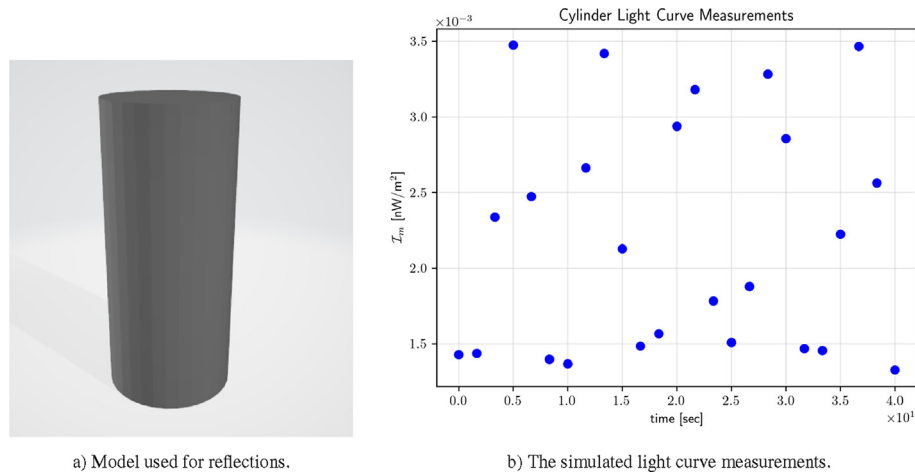


Fig. 3. The cylinder model and measurements for the Case 1 object.

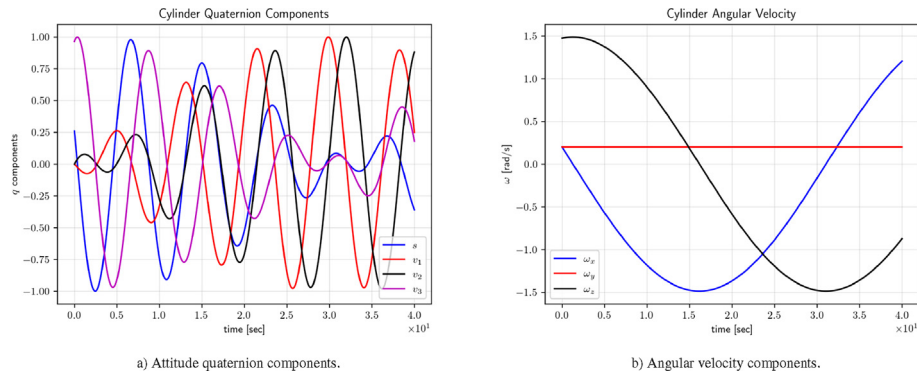


Fig. 4. The cylinder's true attitude and angular velocity time history.

Table 4

Errors in the Case 1 cylinder axis pointing direction and inertial frame angular momentum vector estimates.

Rank	E_{frac}	Axis Estimate Error [deg]		Angular Momentum Error	
		Initial	Average	Magnitude [%]	Direction [deg]
1	$2.426 \cdot 10^{-5}$	129.291	91.272	$-4.861 \cdot 10^{-3}$	$180 - 0.5936$
2	$9.961 \cdot 10^{-5}$	50.711	88.736	$-4.689 \cdot 10^{-3}$	$180 - 0.5819$
3	$1.126 \cdot 10^{-4}$	13.381	9.823	$1.296 \cdot 10^{-4}$	13.191
4	$1.870 \cdot 10^{-4}$	131.333	94.000	$-5.154 \cdot 10^{-3}$	166.568
5	$2.455 \cdot 10^{-4}$	166.618	170.236	$1.420 \cdot 10^{-4}$	13.058

measurement time. Only the error in the estimated direction of the cylinder's axis is reported because rotations about the axis of a completely symmetric cylinder would not affect the observed brightness I_m . For similar reasons, the error in the inertia-frame angular momentum vector is used instead of the body-frame angular velocity error. The bold Rank 1 estimate is the one that best fits the input measurements and is, therefore, most likely to be the actual object state, according to the estimator.

The best-fitting estimate for the cylinder case has an axis pointing error of 91.2° on average. The estimate's angular momentum vector has a magnitude within 0.005% of the truth and is rotated nearly 180° away from it. Despite the axis direction errors, the error fraction E_{frac} of the esti-

mate's light curve is only $2.4 \cdot 10^{-5}$. The axis error is likely due to unobservable ambiguities introduced by the object's shape and rotational motion. The true rotation's angular momentum vector is 87.3° away from the object-observer $\hat{u}$ vector. As a result, the observer is viewing the cylinder's end-over-end tumble edge-on, which may be responsible for ambiguities in the attitude estimate beyond what is usually present in the light curve inversion problem.

The Rank 2 estimate in Table 4 has similar characteristics to the Rank 1 estimate. The remaining three estimates all have higher amounts of error in their light curves and are related by various symmetries. The Rank 2 and 4 estimates both have angular momentum vectors pointing about 13° from the truth, while the Rank 3 estimate's

angular momentum vector has a direction error of 167° indicating that it is likely a reflection of the Rank 2 and 4 vectors. In addition, the Rank 2 estimate's axis has a pointing direction error of about 10° and the Rank 4 estimate's axis has an error of about 170° . This fact, combined with their near-identical angular momentum vector errors, makes it likely that the main difference between the two is that the Rank 4 estimate's axis of symmetry is reversed. It is of note that the angular momentum vector for all five listed estimates is very close to the true magnitude.

Despite the ambiguities introduced by the cylinder's symmetry, the PSO attitude estimator is able to identify candidates for the true attitude time history that match the measurements well. Although there is significant error in the direction of the cylinder's axis, the angular momentum vector is estimated accurately within the limits of symmetries in the problem. Additional analysis is required to test the hypothesis that the observation geometry contributed additional ambiguity to the estimation process.

5.3. Case 2: GEO uniformly-reflecting regular tetrahedron

The second test case involves a uniformly-reflecting regular tetrahedron is shown in Fig. 5a with side lengths of 1.63 m. The observed brightness measurements in Fig. 5b are simulated using a Cook-Torrance reflection model (Duvenhage et al., 2013) with a 0.8 diffuse reflection coefficient and a 0.2 specular reflection coefficient for each surface. The roughness parameter is 10. The mass of the tetrahedron is not evenly distributed, and it has the following moments of inertia: $I_{x,\text{tet}} = 1.0 \text{ kg} \cdot \text{m}^2$, $I_{y,\text{tet}} = 1.5 \text{ kg} \cdot \text{m}^2$, $I_{z,\text{tet}} = 2.0 \text{ kg} \cdot \text{m}^2$.

The initial state of the tetrahedron is randomly generated with the magnitude of the angular velocity vector set to 1.5 rad/s.

$$\begin{aligned} \mathbf{q}_0 &= (0.1005, [0.3015 \ 0.5025 \ 0.8040]^T) \\ \boldsymbol{\omega}_0 &= [0.8566 \ -1.1271 \ 0.4959]^T \text{ rad/s} \end{aligned} \quad (40)$$

These initial states correspond to the true attitude time history shown in Fig. 6a where each line corresponds to a

component of the attitude quaternion $\mathbf{q}$. The true angular velocity of the tetrahedron over time is shown in Fig. 6b. Because the tetrahedron is not axisymmetric, the angular velocity components do not follow strictly sinusoidal patterns, and the quaternion components sometimes change abruptly, for example, at around 25 s in Fig. 6a.

The PSO attitude inversion process attempts to find the true attitude time history using only the light curve measurements in Fig. 5b. The possible attitude nsPSO finds 139 attitudes after running for ten iterations with 17,576 particles. The full state PSO generates 59,631 particles by pairing the attitudes with 429 sampled angular velocity guesses. After 50 iterations, the 250 best-fitting attitude estimates are refined by the quasi-Newton method. The complete inversion process takes about 1.5 h to complete. Table 5 shows the best of these automatically refined estimates. For Cases 2 and 3, the total difference between the true and estimated attitudes is reported.

The Rank 1 through 7 estimates are all approximately equal to the true attitude time history. The Rank 1 estimate has a mean attitude error of only 0.3 arcsec and a mean angular velocity error of 0.3 arcsec/s. This low level of error indicates that an asymmetric moment of inertia is sufficient for the true attitude of the object to be observable, even if, visually, it is completely symmetric. The estimated light curve and its normalized errors at each measurement time are shown in Fig. 7. The errors in Fig. 7b are normalized using the magnitude of the observed measurement vector $\|\mathbf{z}\|$. The error in the estimated light curve at each measurement time is at most $(5.409 \cdot 10^{-7}) \cdot \|\mathbf{z}\|$, for a total E_{frac} of $1.225 \cdot 10^{-6}$.

The Rank 8 through 23 estimates are an average of 116.4° away from the true attitude time history and have an average angular velocity error of 2.803 rad/s. These attitude time histories generally have an E_{frac} of about $1.7 \cdot 10^{-5}$ and likely fit the measurements so well because of the symmetries in the object shape. However, their E_{frac} value is an order of magnitude higher than the Rank 1 estimate, meaning that the accurate attitude estimates are still identifiable for noiseless measurements.

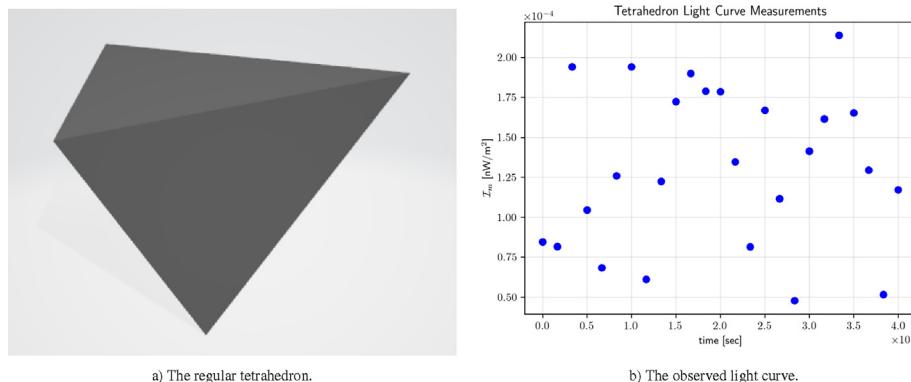


Fig. 5. The Case 2 observed object and measured light curve (Burton et al., 2024).

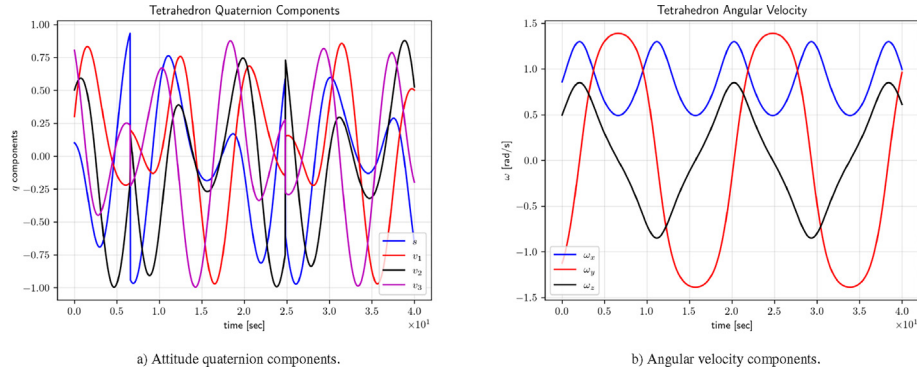


Fig. 6. The tetrahedron's true attitude and angular velocity time history.

Table 5

Errors in the Case 2 tetrahedron attitude and angular velocity estimates.

Rank	E_{frac}	Initial Estimate Error		Average Estimate Error	
		Attitude [deg]	Ang. Vel. [rad/s]	Attitude [deg]	Ang. Vel. [rad/s]
1	$1.225 \cdot 10^{-6}$	$1.005 \cdot 10^{-4}$	$2.071 \cdot 10^{-6}$	$7.841 \cdot 10^{-5}$	$1.340 \cdot 10^{-6}$
...					
8	$1.665 \cdot 10^{-5}$	157.250	2.831	116.388	2.803
...					
24	$8.233 \cdot 10^{-4}$	$180 - 0.02302$	$7.463 \cdot 10^{-4}$	$180 - 0.06394$	$3.839 \cdot 10^{-4}$

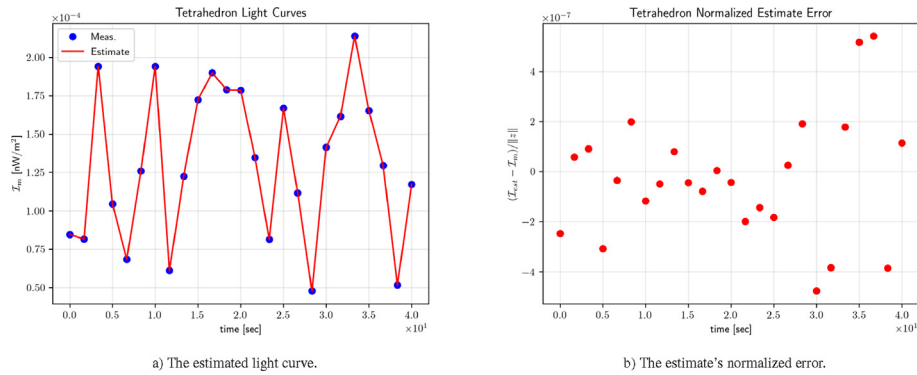


Fig. 7. The best estimated light curve and its normalized error at each measurement time for Case 2.

The Rank 24 through 33 estimates are approximations of the “flipped” attitude time history discussed in Section 4. Note that a completely “flipped” attitude time history for this light curve would have an E_{frac} of $2.0 \cdot 10^{-3}$, higher than any of the estimates listed in Table 5. The Rank 24 through 33 estimates are “nearly-flipped” attitude time histories with a lower E_{frac} . However, as was the case for the Rank 8 through 23 estimates, these E_{frac} values are orders of magnitude higher than the accurate attitude estimate's E_{frac} , meaning that the two can be distinguished.

5.4. Case 3: Full satellite model

The third test case involves the same attitude profile and problem geometry as Case 2, but the observed object's shape is changed to that of the Landsat 8 satellite (Meaney, 2016) as shown in Fig. 8a. This test case aims

to examine the effect of an object's shape and reflection model on the attitude estimate. Hence, the new object retains the same initial conditions as in (40), and its true attitude time history is identical to that shown in Fig. 6.

As in Case 2, the model reflects light according to a Cook-Torrance model (Duvenhage et al., 2013) with a 0.8 diffuse reflection coefficient, a 0.2 specular reflection coefficient, and a roughness parameter of 10 on each surface. However, because the shape of the Landsat 8 model is complex and self-shadowing must be taken into account, accurately simulating reflections for the millions of measurements generated by the PSOs is extremely computationally intensive. Therefore, the reflections are approximated using a neural network to reduce computation time.

The network is trained using one million measurements generated using a high-quality simulator. The network

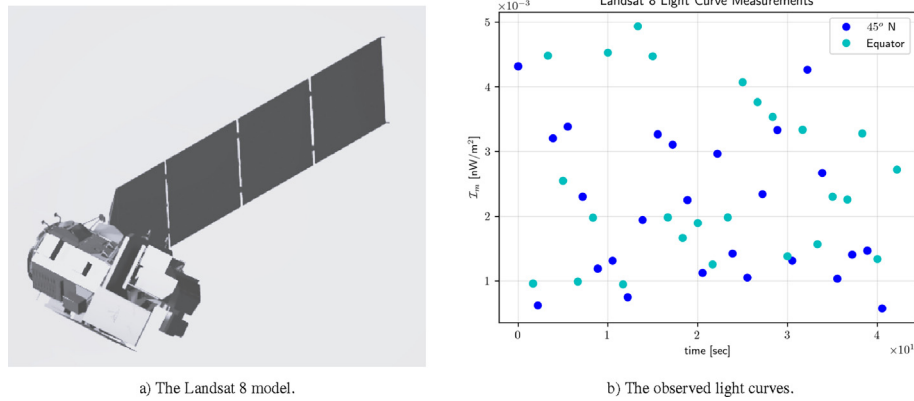


Fig. 8. The Landsat 8 object model and two-observer light curve for Case 3 (Meaney, 2016; Burton et al., 2024).

takes the body-frame components of the $\hat{u}$ and $\hat{s}$ vectors as inputs and outputs the measurement as a floating point number. It has three fully connected convolutional hidden layers with 150 nodes in the first and third hidden layers and 50 nodes in the second layer. The nodes' activation function is ReLu (Nijhawan and Dhar Choudhury, 1999), and the network's weights are optimized using stochastic gradient descent using Adam (Kingma and Ba, 2017). The resulting neural network is used to produce both the observed measurements $\mathbf{z}$ and the estimated measurements $\hat{\mathbf{z}}$.

Using this approximation for the reflection model results in decreased estimate accuracy, even though the same model is used both to compute the measurements and by the estimator. If a similarly-trained neural network is used to approximate the light curve for the tetrahedron in Case 2, then the attitude estimate that best fits the measured light curves has an E_{frac} of 0.1096 compared to an E_{frac} of $1.225 \cdot 10^{-6}$ for the actual Case 2. This estimate has an average attitude error of 115.2° , compared to an error of $(1.34 \cdot 10^{-6})^\circ$ for the original test case.

This increased estimate error caused by the reflection approximations can be compensated for by using multiple observers to increase the attitude's observability. Case 3 uses two near-simultaneous light curves measured by two observers to find the attitude estimate of the Landsat 8 model. The first light curve is measured by an observer at 45° N latitude, as in the previous two test cases. The second light curve is offset by 2.2 s relative to the first and is measured from a point directly south of the first observer on the equator.

The possible attitude nsPSO is run for ten iterations using 314,432 particles and finds 227 possible attitudes. These initial attitudes are then given to the full state PSO. The second PSO is run for 50 iterations, and the 250 best state estimates and their “flipped” counterparts are refined using the quasi-Newton method. Unlike Cases 1 and 2, the epoch time used by the two PSOs is the final measurement taken by the 45° N observer. This time is chosen because it corresponds to the dimmest measurements in the light curve and, therefore, restricts the space of possible attitudes more than the first measurement. The complete inversion process takes about 5.6 h to run.

The five estimates that best fit the input light curves are shown in Table 6. The Rank 1 estimate has an E_{frac} of only $7.617 \cdot 10^{-3}$ and an average attitude error of 0.5905° . The light curves corresponding to the Rank 1 estimate, as seen by the two observers, are shown in Fig. 9. From the plot of measurement errors in Fig. 9b, the largest normalized error in any single estimated measurement is $(3.264 \cdot 10^{-3}) \cdot \|\mathbf{z}\|$.

The Rank 2 through 5 estimates all have an average attitude error that ranges between 2.322° and 9.565° . These estimates also have low errors in their angular velocity estimates, never exceeding an average error of 0.03856 rad/s out of a true angular velocity magnitude of 1.5 rad/s.

Unlike the single-observer attitude estimates for the tetrahedron, the two-observer estimates reported in Table 6 do not include any “flipped” attitude time histories. “Flipped” attitude time histories are not present because they cannot easily fit light curves measured by multiple observers simultaneously. Therefore, having multiple observers may be useful when estimating attitude using real

Table 6
Errors in the Case 3 Landsat 8 attitude and angular velocity estimates.

Rank	E_{frac}	Initial Estimate Error		Average Estimate Error	
		Attitude [deg]	Ang. Vel. [rad/s]	Attitude [deg]	Ang. Vel. [rad/s]
1	$7.617 \cdot 10^{-3}$	0.7852	0.01386	0.5905	$7.139 \cdot 10^{-3}$
2	0.01222	4.389	$6.782 \cdot 10^{-3}$	4.616	$3.677 \cdot 10^{-3}$
3	0.01558	4.449	$5.125 \cdot 10^{-3}$	4.486	$3.075 \cdot 10^{-3}$
4	0.02230	1.979	0.07731	2.322	0.03856
5	0.02676	9.178	$4.774 \cdot 10^{-3}$	9.565	$7.117 \cdot 10^{-3}$

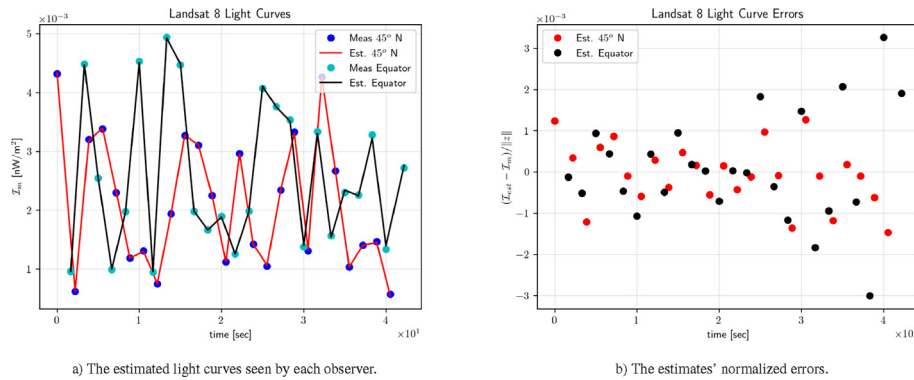


Fig. 9. The best estimated light curves and their normalized errors at each measurement time for Case 3.

measurements whose noise might otherwise make it difficult to distinguish accurate attitude estimates from “flipped” ones.

Further testing also shows that the accuracy and consistency of the attitude estimator for this case varies depending on the observed object’s initial attitude and angular velocity. Therefore, additional work must be done to tune the estimator properly for cases involving approximated measurements. Since more accurate estimates are correlated with lower E_{frac} values, such tuning can be done using only the light curve measurements, even if the true state of the object is unknown.

6. Discussion of potential application to real data

The simulated light curve test cases in Section 5 demonstrate the ability of the method to account for inherent ambiguities present in the light curve inversion problem. Future applications of the method to real measurements will need to address the challenges of uncertainties in the object’s shape and reflection properties, among other areas. In particular, differences between the observed object’s true shape and the inversion method’s shape model have been shown to degrade the quality of the attitude inversion method’s output. Similarly, incorrect parameters in the object’s reflection model can lead to large discrepancies between the observed and estimated light curve. Appending simple shape and reflection model parameters as additional states to the PSOs is one possible way to account for these uncertainties (Burton, 2024). The effect of measurement noise on the attitude inversion process must also be investigated and accounted for. Incorporating statistical measures into the cost functions in (28) and (35) may be one way of accounting for measurement noise.

7. Conclusions

Artificial space objects may have stable or tumbling attitude profiles and cannot be assumed to rotate at a constant rate about one axis. Similarly, when performing a light curve inversion, there is not always any *a priori* information about an observed object’s attitude state. The particle

swarm optimizer (PSO) inversion method addresses these difficulties by estimating an object’s attitude and angular velocity using only its light curve without needing any *a priori* information about the object’s rotation or an initial guess for its state. Arbitrary torque-free attitude motion can be quickly simulated using analytic expressions for attitude. The PSO inversion method only assumes that the problem geometry and the object’s shape and reflective properties are known. Using PSOs to search the space of possible attitude and angular velocity states allows the method to account for the ambiguity of the estimation problem.

The results of the inversion process are shown for three test cases using noiseless simulated measurements. In the first test case, multiple attitude time histories are found that closely match 25 simulated measurements of a tumbling cylinder in geosynchronous equatorial orbit. Symmetries in the object and its tumbling attitude profile may contribute to the attitude ambiguity.

In the second test case, the correct attitude time history is found for a visually symmetric regular tetrahedron. The solver can identify the attitude time histories closest to the truth because of the asymmetries in the tetrahedron’s moments of inertia.

The third test case applies the attitude inversion process to a more complex Landsat 8 satellite model using near-simultaneous measurements from two observers at different locations. A neural network approximates a Cook-Torrance reflection model, making the attitude estimation process computationally tractable. The best estimate found by the PSO method for the Landsat 8 model has an average attitude error of less than 1°. This performance is not consistent for all initial states, however. Additional tuning of the solver or a more robust reflection approximation method may improve the consistency of the attitude estimate for complex non-convex shapes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Attitude representation derivation

The expressions for $\Delta \mathbf{q}_{\text{SAM}}$ and $\Delta \mathbf{q}_{\text{LAM}}$ in (6) and (7) are derived. First, the quaternions $\mathbf{q}_{313}$ and $\mathbf{q}_{131}$ relating Euler angles to a quaternion are derived, then the quaternions are used to describe the attitude of the rotating object. Finally, the initial Euler angle expressions (23), (24), and (25) are derived.

A.1. 313 Sequence quaternion

The rotation corresponding to a 313 (ϕ, θ, ψ) Euler angle sequence is written in quaternion form as follows:

$$\mathbf{q}_{313} = (s_{313}, \mathbf{v}_{313}) = \mathbf{q}_3(\psi) \otimes \mathbf{q}_1(\theta) \otimes \mathbf{q}_3(\phi) \quad (\text{A.1})$$

where $\mathbf{q}_1(\vartheta)$ or $\mathbf{q}_3(\vartheta)$ represents rotation about a single axis by an angle ϑ .

$$\mathbf{q}_{131}(\phi, \theta, \psi) = \left(\cos \frac{\phi}{2} \cos \frac{\theta}{2} \cos \frac{\psi}{2} - \sin \frac{\phi}{2} \cos \frac{\theta}{2} \sin \frac{\psi}{2}, \begin{bmatrix} \sin \frac{\phi}{2} \cos \frac{\theta}{2} \cos \frac{\psi}{2} + \cos \frac{\phi}{2} \cos \frac{\theta}{2} \sin \frac{\psi}{2} \\ \cos \frac{\phi}{2} \sin \frac{\theta}{2} \sin \frac{\psi}{2} - \sin \frac{\phi}{2} \sin \frac{\theta}{2} \cos \frac{\psi}{2} \\ \cos \frac{\phi}{2} \sin \frac{\theta}{2} \cos \frac{\psi}{2} + \sin \frac{\phi}{2} \sin \frac{\theta}{2} \sin \frac{\psi}{2} \end{bmatrix} \right) \quad (\text{A.7})$$

$$\begin{aligned} \mathbf{q}_1(\vartheta) &= \left(\cos \frac{\vartheta}{2}, [\sin \frac{\vartheta}{2} \ 0 \ 0]^T \right) \\ \mathbf{q}_3(\vartheta) &= \left(\cos \frac{\vartheta}{2}, [0 \ 0 \ \sin \frac{\vartheta}{2}]^T \right) \end{aligned} \quad (\text{A.2})$$

The quaternion multiplication $\otimes$ is defined as follows (Markley and Crassidis, 2014):

$$\begin{aligned} \mathbf{q}_C &= (s_C, \mathbf{v}_C) = \mathbf{q}_A \otimes \mathbf{q}_B = (s_A, \mathbf{v}_A) \otimes (s_B, \mathbf{v}_B) \\ s_C &= s_A s_B - \mathbf{v}_A \cdot \mathbf{v}_B \\ \mathbf{v}_C &= s_A \mathbf{v}_B + s_B \mathbf{v}_A - \mathbf{v}_A \times \mathbf{v}_B \end{aligned} \quad (\text{A.3})$$

Multiplying the three quaternions in (A.1) results in the following expression for $\mathbf{q}_{313}$:

$$\mathbf{q}_{313}(\phi, \theta, \psi) = \left(\cos \frac{\phi}{2} \cos \frac{\theta}{2} \cos \frac{\psi}{2} - \sin \frac{\phi}{2} \cos \frac{\theta}{2} \sin \frac{\psi}{2}, \begin{bmatrix} \cos \frac{\phi}{2} \sin \frac{\theta}{2} \cos \frac{\psi}{2} + \sin \frac{\phi}{2} \sin \frac{\theta}{2} \sin \frac{\psi}{2} \\ \sin \frac{\phi}{2} \sin \frac{\theta}{2} \cos \frac{\psi}{2} - \cos \frac{\phi}{2} \sin \frac{\theta}{2} \sin \frac{\psi}{2} \\ \cos \frac{\phi}{2} \cos \frac{\theta}{2} \sin \frac{\psi}{2} + \sin \frac{\phi}{2} \cos \frac{\theta}{2} \cos \frac{\psi}{2} \end{bmatrix} \right) \quad (\text{A.4})$$

The expression in (A.4) can be further simplified using the sum and difference trigonometric identities (Falduto et al., 2015) to reach the form shown below:

$$\mathbf{q}_{313}(\phi, \theta, \psi) = \left(\cos \frac{\theta}{2} \cos \left(\frac{\phi + \psi}{2} \right), \begin{bmatrix} \sin \frac{\theta}{2} \cos \left(\frac{\phi - \psi}{2} \right) \\ \sin \frac{\theta}{2} \sin \left(\frac{\phi - \psi}{2} \right) \\ \cos \frac{\theta}{2} \sin \left(\frac{\phi + \psi}{2} \right) \end{bmatrix} \right) \quad (\text{A.5})$$

A.2. 131 Sequence quaternion

The quaternion $\mathbf{q}_{131}(\phi, \theta, \psi)$ corresponding to a 131 (ϕ, θ, ψ) Euler angle sequence is calculated in a similar way to the 313 sequence quaternion. The quaternion $\mathbf{q}_{131}(\phi, \theta, \psi)$ is equivalent to the following product:

$$\mathbf{q}_{131}(\phi, \theta, \psi) = \mathbf{q}_1(\psi) \otimes \mathbf{q}_3(\theta) \otimes \mathbf{q}_1(\phi) \quad (\text{A.6})$$

where $\mathbf{q}_i(\vartheta)$ are defined as in (A.2) and $\otimes$ is defined as in (A.3). The product of the three quaternions is shown below:

Simplifying (A.7) with the sum and difference trigonometric identities (Falduto et al., 2015) results in the following expression:

$$\mathbf{q}_{131}(\phi, \theta, \psi) = \left(\cos \frac{\theta}{2} \cos \left(\frac{\phi + \psi}{2} \right), \begin{bmatrix} \cos \frac{\theta}{2} \sin \left(\frac{\phi + \psi}{2} \right) \\ -\sin \frac{\theta}{2} \sin \left(\frac{\phi - \psi}{2} \right) \\ \sin \frac{\theta}{2} \cos \left(\frac{\phi - \psi}{2} \right) \end{bmatrix} \right) \quad (\text{A.8})$$

A.3. Short-axis mode attitude representation

In SAM rotation, the object is rotating principally about its short axis $\hat{b}_z$. Therefore, in a 313 (ϕ, θ, ψ) Euler angle sequence, the first angle ϕ is the precession angle of the object about the angular momentum unit vector $\hat{h}$, the second angle θ is the nutation angle between $\hat{h}$ and $\hat{b}_z$, and the third angle ψ is the spin angle about $\hat{b}_z$.

Substituting (A.12)–(A.14) into (A.10) results in the final expression for $\mathbf{q}_{H,SAM}$ for SAM rotations:

$$\mathbf{q}_{H,SAM} = \left(\sqrt{\frac{h_z + 1}{2}}, \frac{1}{\sqrt{2(h_z + 1)}} [-h_y \ h_x \ 0]^T \right) \quad (\text{A.15})$$

Multiplying $\mathbf{q}_{313}$ by $\mathbf{q}_{H,SAM}$ and taking the conjugate results in the following expression for attitude:

$$\Delta \mathbf{q}_{SAM}(\phi, \theta, \psi) = \frac{1}{\sqrt{2(h_z + 1)}} \left((h_z + 1) \cos \frac{\theta}{2} \cos \left(\frac{\phi + \psi}{2} \right) + \sin \frac{\theta}{2} (h_y \cos \left(\frac{\phi - \psi}{2} \right) - h_x \sin \left(\frac{\phi - \psi}{2} \right)), \right. \\ \left. \begin{bmatrix} -(h_z + 1) \sin \frac{\theta}{2} \cos \left(\frac{\phi - \psi}{2} \right) - \cos \frac{\theta}{2} (h_x \sin \left(\frac{\phi + \psi}{2} \right) - h_y \cos \left(\frac{\phi + \psi}{2} \right)) \\ -(h_z + 1) \sin \frac{\theta}{2} \sin \left(\frac{\phi - \psi}{2} \right) - \cos \frac{\theta}{2} (h_x \cos \left(\frac{\phi + \psi}{2} \right) + h_y \sin \left(\frac{\phi + \psi}{2} \right)) \\ -(h_z + 1) \cos \frac{\theta}{2} \sin \left(\frac{\phi + \psi}{2} \right) + \sin \frac{\theta}{2} (h_x \cos \left(\frac{\phi - \psi}{2} \right) + h_y \sin \left(\frac{\phi - \psi}{2} \right)) \end{bmatrix} \right) \quad (\text{A.16})$$

The attitude of an object undergoing SAM rotation can be represented by first making a rotation to align $\hat{b}_z$ with $\hat{h}$, then multiplying by $\mathbf{q}_{313}$.

$$\Delta \mathbf{q}_{SAM} = (\mathbf{q}_{313}(\phi, \theta, \psi) \otimes \mathbf{q}_{H,SAM})^* \quad (\text{A.9})$$

where $\mathbf{q}_{H,SAM}$ corresponds to a rotation that aligns $\hat{b}_z$ and $\hat{h}$. Since $\mathbf{q}_{313}$ and $\mathbf{q}_{H,SAM}$ rotate from the body-fixed frame to the inertial frame, their conjugate must be taken before they can be used to represent attitude with $\mathbf{q}_{SAM}$.

The quaternion $\mathbf{q}_{H,SAM}$ is written as follows:

$$\mathbf{q}_{H,SAM} = \left(\cos \frac{\theta_H}{2}, \sin \frac{\theta_H}{2} \frac{\hat{b}_z \times \hat{h}}{|\hat{b}_z \times \hat{h}|} \right) \quad (\text{A.10})$$

where θ_H is the angle between $\hat{h}$ and $\hat{b}_z$. If, at the initial time, the angular momentum unit vector written using body-frame coordinates is equal to $\hat{h} = [h_x \ h_y \ h_z]^T$, then

$$\cos \theta_H = \hat{h} \cdot \hat{b}_z = h_z \quad (\text{A.11})$$

The scalar component of (A.10) is found using the cosine half-angle formula.

$$\cos \frac{\theta_H}{2} = \sqrt{\frac{\cos \theta_H + 1}{2}} = \sqrt{\frac{h_z + 1}{2}} \quad (\text{A.12})$$

The factor $\sin(\theta_H/2)$ is found in a similar way:

$$\sin \frac{\theta_H}{2} = \sqrt{\frac{1 - \cos \theta_H}{2}} = \sqrt{\frac{1 - h_z}{2}} \quad (\text{A.13})$$

The magnitude of the cross-product is:

$$|\hat{b}_z \times \hat{h}| = \sqrt{h_x^2 + h_y^2} = \sqrt{1 - h_z^2} \\ = \sqrt{(1 - h_z)(1 + h_z)} \quad (\text{A.14})$$

If the initial Euler angles are chosen such that (A.16) is equal to the identity quaternion $(1, \mathbf{0})$, then (A.16) represents the change in attitude from the initial time step, and the SAM attitude representation can be written as follows:

$$\mathbf{q}_{SAM} = \Delta \mathbf{q}_{SAM} \otimes \mathbf{q}_0 \quad (\text{A.17})$$

where $\Delta \mathbf{q}_{SAM}$ is as in (6) and $\mathbf{q}_0$ is the attitude quaternion at the initial time step.

A.4. Long-axis mode attitude representation

The expression for LAM attitude change $\Delta \mathbf{q}_{LAM}$ is derived in a similar way to $\Delta \mathbf{q}_{SAM}$. For LAM motion, the three angles of a 131 (ϕ, θ, ψ) Euler angle sequence correspond to the precession ϕ , nutation θ , and spin ψ angles of an object relative to the $\hat{h}$ as it rotates primarily about the body-fixed $\hat{b}_x$ axis. The quaternion $\Delta \mathbf{q}_{LAM}$ is therefore equal to the following:

$$\Delta \mathbf{q}_{LAM} = (\mathbf{q}_{131}(\phi, \theta, \psi) \otimes \mathbf{q}_{H,LAM})^* \quad (\text{A.18})$$

where $\mathbf{q}_{H,LAM}$ represents a rotation aligning $\hat{b}_x$ with $\hat{h}$. It is found using a similar method to $\mathbf{q}_{H,SAM}$.

$$\mathbf{q}_{H,LAM} = \left(\sqrt{\frac{h_x + 1}{2}}, \frac{1}{\sqrt{2(h_x + 1)}} [0 \ -h_z \ h_y]^T \right) \quad (\text{A.19})$$

The product in (A.18) is then used to calculate $\Delta \mathbf{q}_{LAM}$.

Choosing the initial Euler angles such that (A.20) is equal

$$\Delta \mathbf{q}_{\text{LAM}}(\phi, \theta, \psi) = \frac{1}{\sqrt{2(h_x+1)}} \left((h_x+1) \cos \frac{\theta}{2} \cos \left(\frac{\phi+\psi}{2} \right) - \sin \frac{\theta}{2} (h_z \sin \left(\frac{\phi-\psi}{2} \right) + h_y \cos \left(\frac{\phi-\psi}{2} \right)) \right. \\ \left. \begin{bmatrix} -(h_x+1) \cos \frac{\theta}{2} \sin \left(\frac{\phi+\psi}{2} \right) + \sin \frac{\theta}{2} (h_z \cos \left(\frac{\phi-\psi}{2} \right) - h_y \sin \left(\frac{\phi-\psi}{2} \right)) \\ (h_x+1) \sin \frac{\theta}{2} \sin \left(\frac{\phi-\psi}{2} \right) + \cos \frac{\theta}{2} (h_z \cos \left(\frac{\phi+\psi}{2} \right) - h_y \sin \left(\frac{\phi+\psi}{2} \right)) \\ -(h_x+1) \sin \frac{\theta}{2} \cos \left(\frac{\phi-\psi}{2} \right) - \cos \frac{\theta}{2} (h_y \cos \left(\frac{\phi+\psi}{2} \right) + h_z \sin \left(\frac{\phi+\psi}{2} \right)) \end{bmatrix} \right) \quad (\text{A.20})$$

to the identity quaternion $(1, \mathbf{0})$ at the initial time t_0 allows $\mathbf{q}_{\text{LAM}}$ to be written as follows:

$$\mathbf{q}_{\text{LAM}} = \Delta \mathbf{q}_{\text{LAM}} \otimes \mathbf{q}_0 \quad (\text{A.21})$$

where $\Delta \mathbf{q}_{\text{LAM}}$ is equal to (7).

A.5. Short-axis mode initial euler angles

At the initial time step, the Euler angles ϕ_0, θ_0 , and ψ_0 are chosen such that $\Delta \mathbf{q}_{\text{SAM}}(t_0) = (1, \mathbf{0})$. The initial nutation angle θ_0 and spin angle ψ_0 for SAM rotations are found using the initial angular momentum unit vector $\hat{\mathbf{h}} = [h_x \ h_y \ h_z]^T$ written in the body-fixed frame. An object is undergoing SAM rotation if the dynamic moment of inertia I_d from (2) is greater than the intermediate moment of inertia I_y . The components of the unit vector $\hat{\mathbf{h}}$ can be written in terms of the 313 Euler angles as follows:

$$h_x = \sin \theta_0 \sin \psi_0 \quad h_y = \sin \theta_0 \cos \psi_0 \quad h_z = \cos \theta_0 \quad (\text{A.22})$$

From (A.22) it is obvious that the initial nutation angle θ_0 is equal to the following:

$$\theta_0 = \cos^{-1}(h_z) \quad (\text{A.23})$$

The above expression is identical to (23) for $I_d > I_y$. The initial nutation angle θ_0 is always chosen such that $\theta_0 \in [0, \pi]$.

The initial value of the spin angle ψ_0 can also be found:

$$\psi_0 = \tan^{-1} \left(\frac{h_z / \sin \theta_0}{h_y / \sin \theta_0} \right) \quad (\text{A.24})$$

This expression is equivalent to (24) for $I_d > I_y$.

Since the precession angle ϕ does not appear in (A.22), the initial precession angle ϕ_0 is found by setting $\Delta \mathbf{q}_{\text{SAM}} = (1, \mathbf{0})$. Since $\Delta \mathbf{q}_{\text{SAM}}$ is a unit quaternion, it will be sufficient to find a ϕ_0 that results in the scalar component of (6) equalling 1.

$$\frac{1}{\sqrt{2(h_z+1)}} \left\{ (h_z+1) \cos \frac{\theta_0}{2} \cos \left(\frac{\phi_0+\psi_0}{2} \right) + \sin \frac{\theta_0}{2} \left[h_y \cos \left(\frac{\phi_0-\psi_0}{2} \right) - h_x \sin \left(\frac{\phi_0-\psi_0}{2} \right) \right] \right\} = 1 \quad (\text{A.25})$$

The sine and cosine sum and difference formulas (Falduto et al., 2015) are used to expand (A.25) so that ϕ_0 can be separated from the other Euler angles.

$$(h_z+1) \cos \frac{\theta_0}{2} (\cos \frac{\phi_0}{2} \cos \frac{\psi_0}{2} - \sin \frac{\phi_0}{2} \sin \frac{\psi_0}{2}) + \sin \frac{\theta_0}{2} [h_y (\cos \frac{\phi_0}{2} \cos \frac{\psi_0}{2} + \sin \frac{\phi_0}{2} \sin \frac{\psi_0}{2}) - h_x (\sin \frac{\phi_0}{2} \cos \frac{\psi_0}{2} - \cos \frac{\phi_0}{2} \sin \frac{\psi_0}{2})] = \sqrt{2(h_z+1)} \quad (\text{A.26})$$

After rearranging the terms, (A.26) is rewritten as an equation in the following form:

$$A \cos \frac{\phi_0}{2} + B \sin \frac{\phi_0}{2} = C \quad (\text{A.27})$$

where A, B , and C are constants:

$$\begin{aligned} A &= (h_z+1) \cos \frac{\theta_0}{2} \cos \frac{\psi_0}{2} + \sin \frac{\theta_0}{2} (h_y \cos \frac{\psi_0}{2} + h_x \sin \frac{\psi_0}{2}) \\ B &= -(h_z+1) \cos \frac{\theta_0}{2} \sin \frac{\psi_0}{2} - \sin \frac{\theta_0}{2} (h_x \cos \frac{\psi_0}{2} - h_y \sin \frac{\psi_0}{2}) \\ C &= \sqrt{2(h_z+1)} \end{aligned} \quad (\text{A.28})$$

The constants in (A.28) can be rewritten solely in terms of θ_0 and ψ_0 using the relations in (A.22).

$$\begin{aligned} A &= (\cos \theta_0 + 1) \cos \frac{\theta_0}{2} \cos \frac{\psi_0}{2} + \sin \theta_0 \sin \frac{\theta_0}{2} (\cos \psi_0 \cos \frac{\psi_0}{2} + \sin \psi_0 \sin \frac{\psi_0}{2}) \\ B &= -(\cos \theta_0 + 1) \cos \frac{\theta_0}{2} \sin \frac{\psi_0}{2} - \sin \theta_0 \sin \frac{\theta_0}{2} (\sin \psi_0 \cos \frac{\psi_0}{2} - \cos \psi_0 \sin \frac{\psi_0}{2}) \\ C &= \sqrt{2(\cos \theta_0 + 1)} \end{aligned} \quad (\text{A.29})$$

The half angles in (A.29) can be further rewritten using the half-angle formulas (Falduto et al., 2015).

$$\cos \frac{\vartheta}{2} = \pm \sqrt{\frac{1 + \cos \vartheta}{2}} \quad \sin \frac{\vartheta}{2} = \pm \sqrt{\frac{1 - \cos \vartheta}{2}} \quad (\text{A.30})$$

where ϑ is some angle. Recalling that $\theta_0 \in [0, \pi]$ and $\psi_0 \in [0, 2\pi]$, it follows that $\cos(\theta_0/2), \sin(\theta_0/2)$, and $\sin(\psi_0/2)$ are always positive, and $\cos(\psi_0/2)$ is negative everywhere that $\sin \psi_0$ is negative. By inspection of (A.29), it can be seen that A will be negative if $\sin \psi_0$ and $\cos(\psi_0/2)$ are negative. It can also be demonstrated that B is always negative, though the proof is omitted.

Simplified forms for A and B in (A.31) are derived by finding expressions for A^2 and B^2 . After simplifying, the following expressions are reached:

$$\begin{aligned} A^2 &= (\cos \theta_0 + 1)(\cos \psi_0 + 1) \quad B^2 \\ &= (\cos \theta_0 + 1)(1 - \cos \psi_0) \end{aligned} \quad (\text{A.32})$$

Note that the same forms are reached whether $\sin \psi_0$ is positive or negative. The constant A from (A.27) can now be written in a form corresponding to (26). The simplified forms of both A and B are given below:

$$\begin{aligned} A &= \pm \left[(\cos \theta_0 + 1) \frac{\sqrt{(1+\cos \theta_0)(1+\cos \psi_0)}}{2} + \sin \theta_0 \frac{\sqrt{1-\cos \theta_0}}{2} (\cos \psi_0 \sqrt{1+\cos \psi_0} + |\sin \psi_0| \sqrt{1-\cos \psi_0}) \right] \\ B &= - \left[(\cos \theta_0 + 1) \frac{\sqrt{(1+\cos \theta_0)(1-\cos \psi_0)}}{2} + \sin \theta_0 \frac{\sqrt{1-\cos \theta_0}}{2} (|\sin \psi_0| \sqrt{1+\cos \psi_0} - \cos \psi_0 \sqrt{1-\cos \psi_0}) \right] \end{aligned} \quad (\text{A.31})$$

$$\begin{aligned} A &= \begin{cases} \sqrt{(\cos \theta_0 + 1)(\cos \psi_0 + 1)} & \text{if } \sin \psi_0 \geq 0 \\ -\sqrt{(\cos \theta_0 + 1)(\cos \psi_0 + 1)} & \text{if } \sin \psi_0 < 0 \end{cases} \\ B &= -\sqrt{(\cos \theta_0 + 1)(1 - \cos \psi_0)} \end{aligned} \quad (\text{A.33})$$

The expression in (A.27) can now be used to find ϕ_0 by performing a Weierstrass substitution (Merlet, 2004) to turn the equation into a quadratic polynomial.

$$\sin \frac{\phi_0}{2} = \frac{2r}{1+r^2} \quad \cos \frac{\phi_0}{2} = \frac{1-r^2}{1+r^2} \quad r = \tan \frac{\phi_0}{4} \quad (\text{A.34})$$

After performing this substitution, it can be shown that (A.27) is equivalent to the following:

$$-(A+C)r^2 + 2Br + (A-C) = 0 \quad (\text{A.35})$$

Since $A+C$ is never zero, the roots of (A.35) can be found using the quadratic formula.

$$\tan \frac{\phi_0}{4} = r = \frac{B \mp \sqrt{A^2 + B^2 - C^2}}{A+C} \quad (\text{A.36})$$

From (26), it is simple to show that $A^2 + B^2 - C^2 = 0$. Therefore, ϕ_0 can be written as follows:

$$\phi_0 = 4 \tan^{-1} \left(\frac{B}{A+C} \right) \quad (\text{A.37})$$

where A and B are defined in (A.33), and C is defined in (A.29). The expression in (A.37) is equivalent to (25).

A.6. Long-axis mode initial angles

The initial Euler angle values for long-axis mode (LAM) rotations are found in a similar way to the SAM initial angles. LAM rotation occurs when the dynamic moment of inertia I_d in (2) is less than the intermediate moment of inertia I_y . For LAM rotation, the initial body-frame angular momentum unit vector $\hat{h} = [h_x \ h_y \ h_z]^T$ components are related to the nutation and spin angles as follows:

$$\begin{aligned} h_x &= \cos \theta_0 \quad h_y = -\sin \theta_0 \cos \psi_0 \quad h_z \\ &= \sin \theta_0 \sin \psi_0 \end{aligned} \quad (\text{A.38})$$

The initial nutation angle θ_0 and spin angle ψ_0 can therefore be written as shown below:

$$\theta_0 = \cos^{-1}(h_x) \quad \psi_0 = \tan^{-1} \left(\frac{h_z / \sin \theta_0}{-h_y / \sin \theta_0} \right) \quad (\text{A.39})$$

The expressions in (A.39) are equivalent to (23) and (24) for $I_d < I_y$.

By setting the scalar component of $\Delta \mathbf{q}_{\text{LAM}}$ in (A.20) equal to 1, it is found that the LAM initial precession angle ϕ_0 relates to the other initial Euler angles using the same relations as found for SAM rotations. The LAM equivalent to (A.27) is as follows:

$$D \cos \frac{\phi_0}{2} + E \sin \frac{\phi_0}{2} = F \quad (\text{A.40})$$

where

$$\begin{aligned} D &= (\cos \theta_0 + 1) \cos \frac{\theta_0}{2} \cos \frac{\psi_0}{2} + \sin \theta_0 \sin \frac{\theta_0}{2} (\cos \psi_0 \cos \frac{\psi_0}{2} + \sin \psi_0 \sin \frac{\psi_0}{2}) \\ E &= -(\cos \theta_0 + 1) \cos \frac{\theta_0}{2} \sin \frac{\psi_0}{2} - \sin \theta_0 \sin \frac{\theta_0}{2} (\sin \psi_0 \cos \frac{\psi_0}{2} - \cos \psi_0 \sin \frac{\psi_0}{2}) \\ F &= \sqrt{2(\cos \theta_0 + 1)} \end{aligned} \quad (\text{A.41})$$

In (A.41), D , E , and F are all identical to the A , B , and C expressions in (A.29). Therefore, the initial precession angle ϕ_0 expressions in (25) and (26) are valid for both SAM and LAM rotations.

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